

SOCIAL MEDIA SENTIMENT ANALYSIS

```
In [2]: from IPython.display import display, Image  
display(Image('social.jpg'))
```



```
In [1]: import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns  
import wordcloud  
import warnings  
warnings.filterwarnings("ignore")  
from wordcloud import WordCloud  
from textblob import TextBlob
```

```
In [7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 732 entries, 0 to 731
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0.1          732 non-null   int64
1   Unnamed: 0            732 non-null   int64
2   Text                  732 non-null   object
3   Sentiment             732 non-null   object
4   Timestamp             732 non-null   object
5   User                  732 non-null   object
6   Platform              732 non-null   object
7   Hashtags              732 non-null   object
8   Retweets              732 non-null   float64
9   Likes                 732 non-null   float64
10  Country               732 non-null   object
11  Year                  732 non-null   int64
12  Month                 732 non-null   int64
13  Day                   732 non-null   int64
14  Hour                  732 non-null   int64
dtypes: float64(2), int64(6), object(7)
memory usage: 85.9+ KB
```

```
In [2]: df=pd.read_csv("sentiment.csv",encoding="utf8")
df.head(2)
```

```
Out[2]:
```

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Has
--	--------------	------------	------	-----------	-----------	------	----------	-----

0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#N #
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#T #Mo



```
In [3]: df['Platform']=df['Platform'].str.strip()
```

```
In [4]: df['Platform']=df['Platform'].str.strip()
```

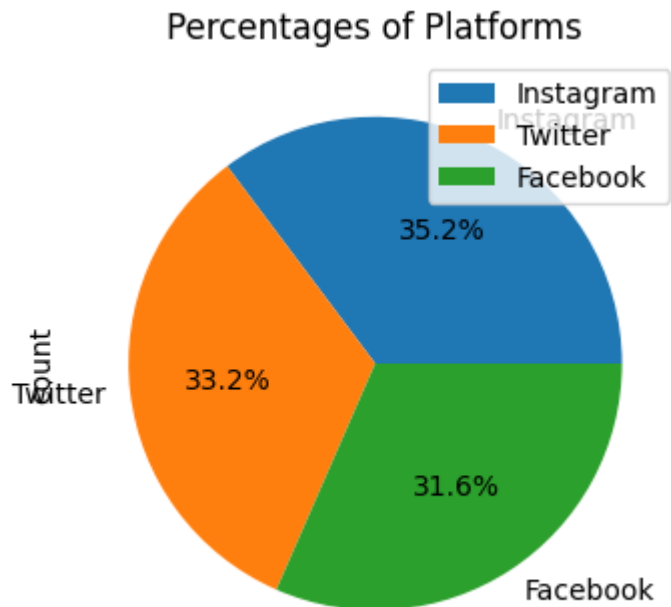
```
In [5]: df['Platform'].value_counts()
```

```
Out[5]: Platform
Instagram    258
Twitter      243
Facebook     231
Name: count, dtype: int64
```

```
In [6]: # Trim Platform
df['Platform']=df['Platform'].str.strip()
```

In []:

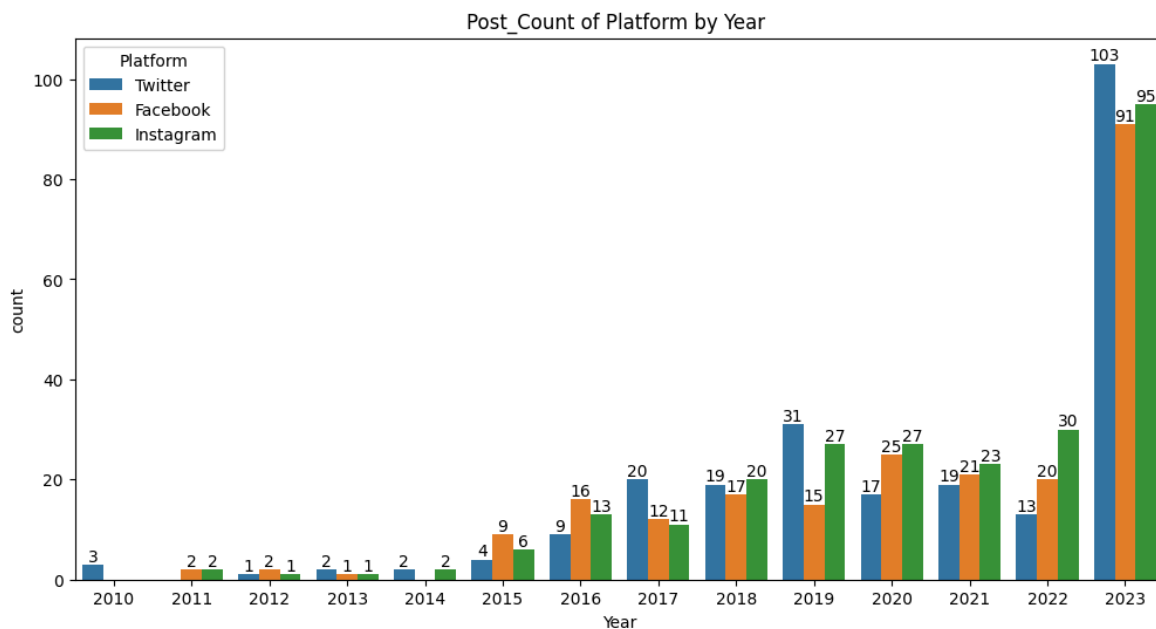
```
In [8]: plt.figure(figsize=(4,4))
df['Platform'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title('Percentages of Platforms')
plt.legend()
plt.show()
```



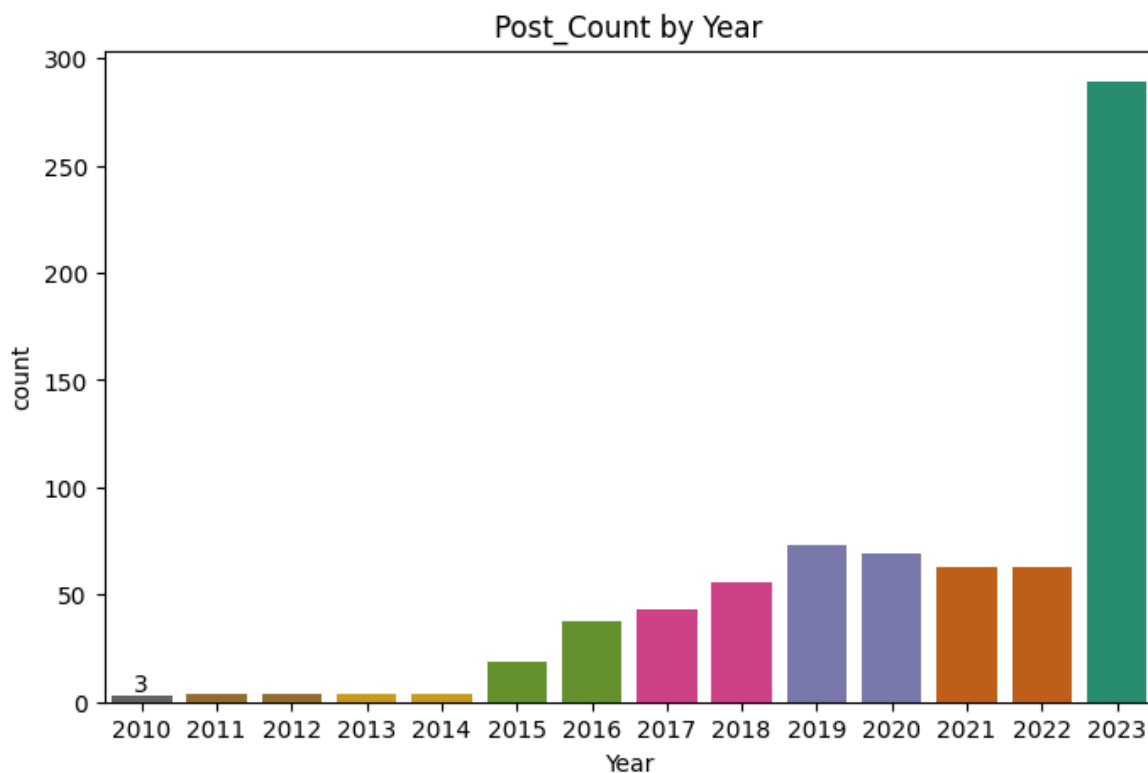
```
In [9]: plt.figure(figsize = (15,6))
ax = sns.countplot(x = "Platform" , data = df ,hue = "Country")
plt.legend(title="Churn", bbox_to_anchor = (1,1))
plt.title("Platform by Country" , fontsize =12 )
plt.show()
```



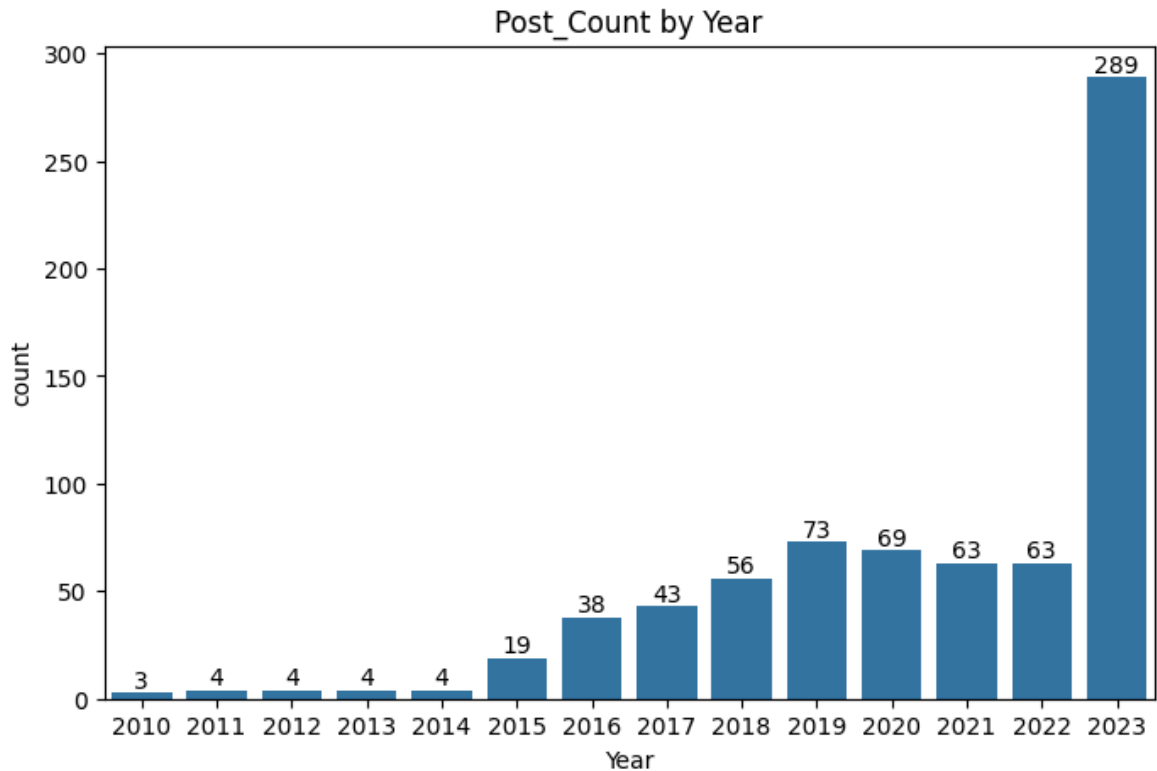
```
In [10]: plt.figure(figsize=(12,6))
ax = sns.countplot(x = "Year" , data = df, hue = "Platform" )
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
ax.bar_label(ax.containers[2])
plt.title("Post_Count of Platform by Year " , fontsize = 12 )
plt.xticks
plt.show()
```



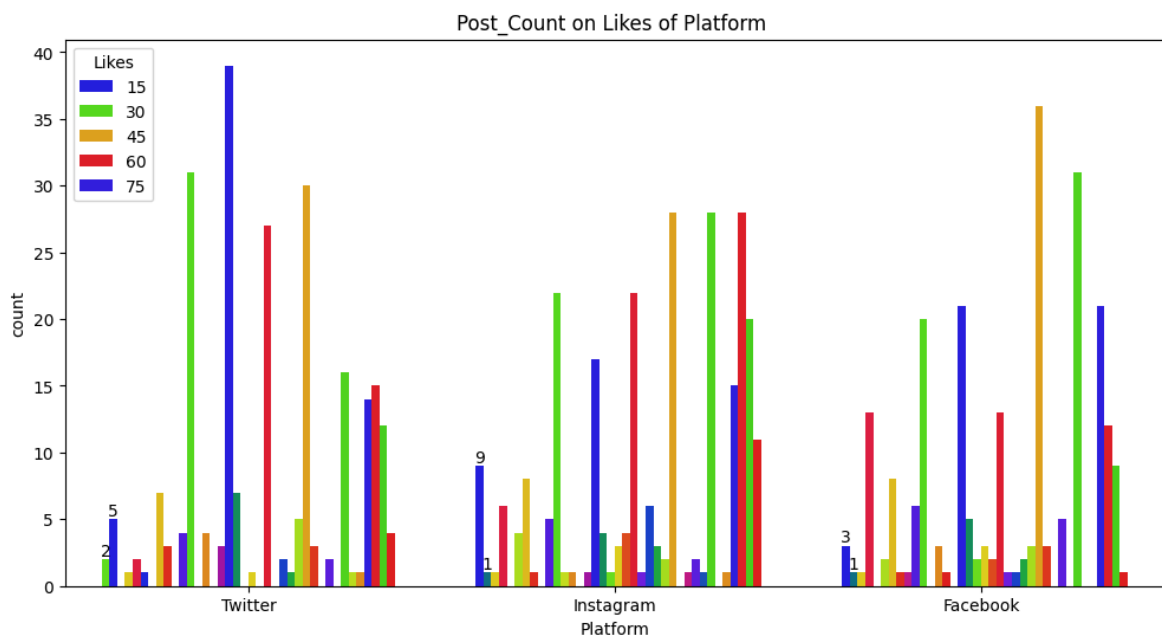
```
In [67]: plt.figure(figsize = (8,5))
ax = sns.countplot(x = "Year" , data = df , palette="Dark2_r")
ax.bar_label(ax.containers[0])
plt.title("Post_Count by Year" , fontsize =12 )
plt.show()
```



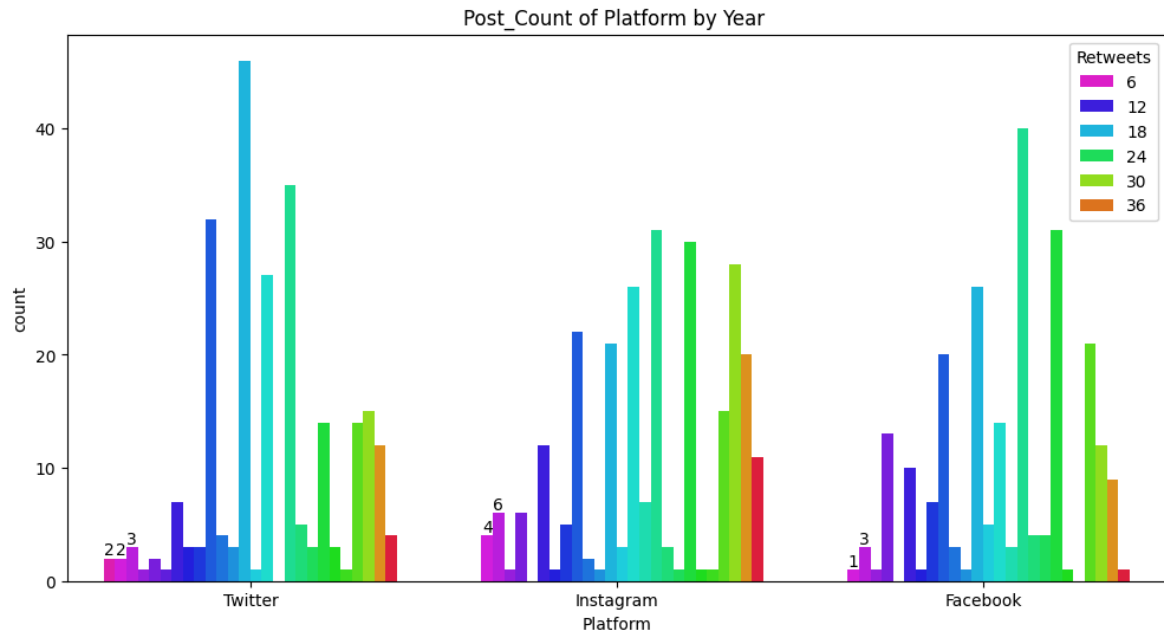
```
In [12]: plt.figure(figsize = (8,5))
ax = sns.countplot(x = "Year" , data =df)
ax.bar_label(ax.containers[0])
plt.title("Post_Count by Year" , fontsize =12 )
plt.show()
```



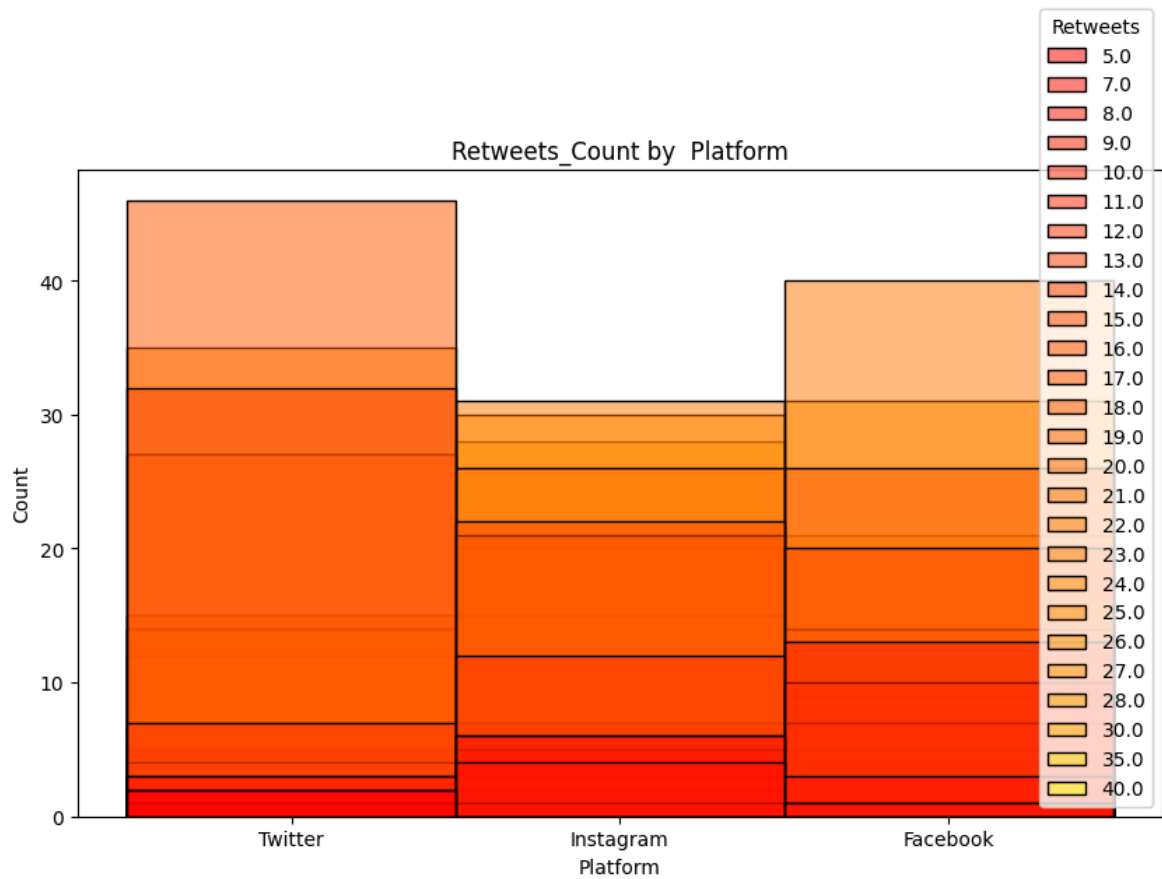
```
In [210]: plt.figure(figsize=(12,6))
ax = sns.countplot(x = "Platform" , data = df, hue = "Likes" ,palette='prism_r')
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
ax.bar_label(ax.containers[2])
plt.title("Post_Count on Likes of Platform " , fontsize = 12 )
plt.xticks
plt.show()
```



```
In [57]: plt.figure(figsize=(12,6))
ax = sns.countplot(x = "Platform" , data = df, hue = "Retweets" ,palette='gist_r
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
ax.bar_label(ax.containers[2])
plt.title("Post_Count of Platform by Year " , fontsize = 12 )
plt.xticks
plt.show()
```



```
In [64]: plt.figure(figsize=(10,6))
ax = sns.histplot(x = "Platform" , data =df ,bins= 50 , hue = "Retweets",palette
plt.title("Retweets_Count by Platform" , fontsize = 12)
plt.show()
```

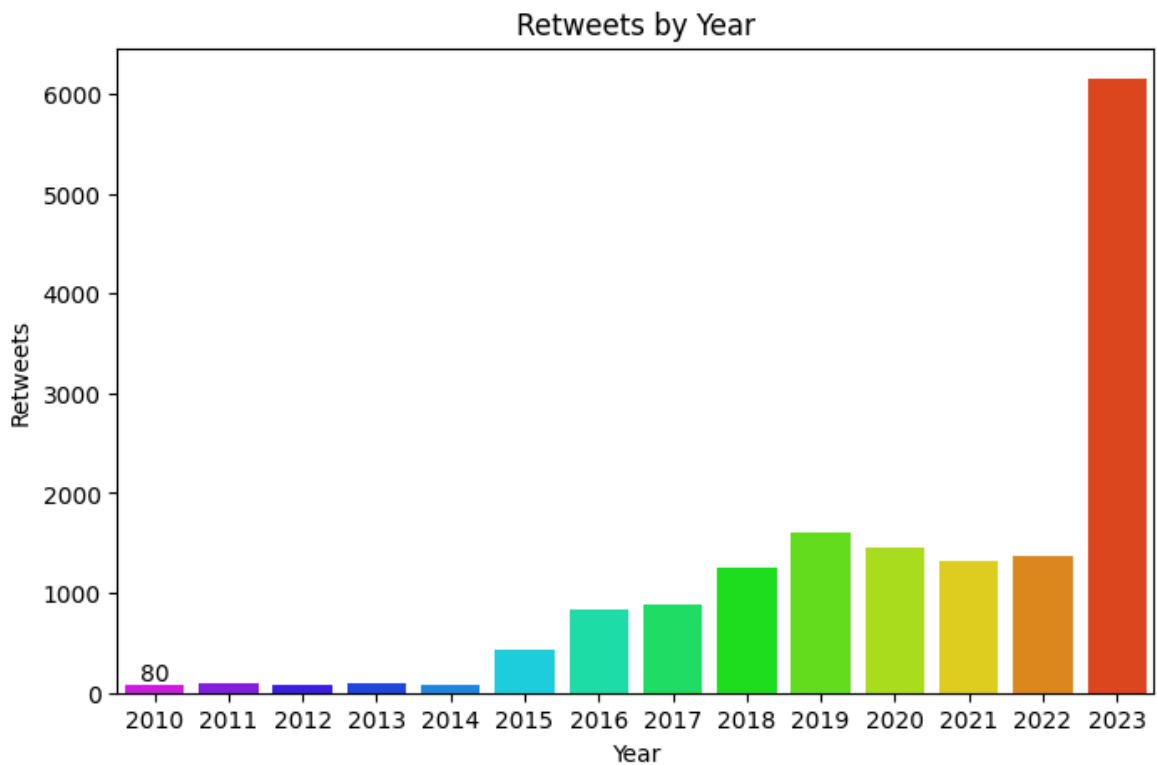


```
In [212... f=df.groupby('Year')['Retweets'].sum().reset_index()
f
```

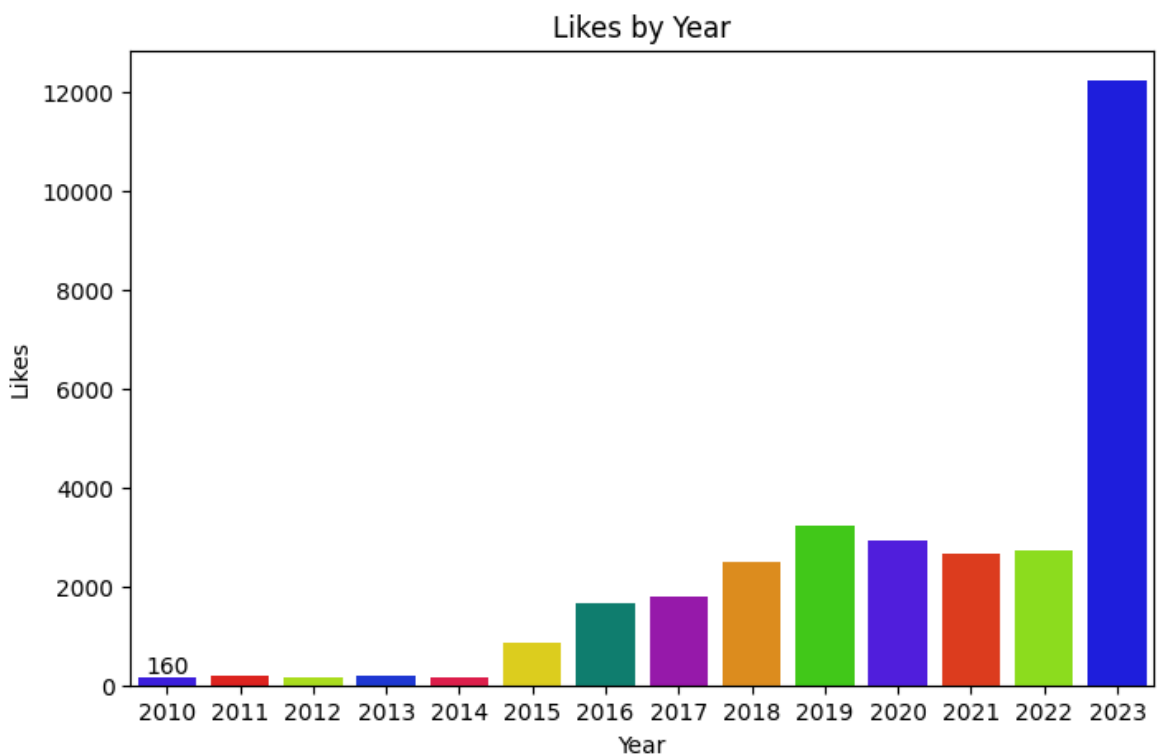
```
Out[212...
   Year  Retweets
0  2010      80.0
1  2011      91.0
2  2012      80.0
3  2013      88.0
4  2014      80.0
5  2015     434.0
6  2016     836.0
7  2017     892.0
8  2018    1253.0
9  2019    1611.0
10 2020    1458.0
11 2021    1326.0
12 2022    1367.0
13 2023    6148.0
```

```
In [214... f=df.groupby('Year')['Retweets'].sum().reset_index()
plt.figure(figsize = (8,5))
```

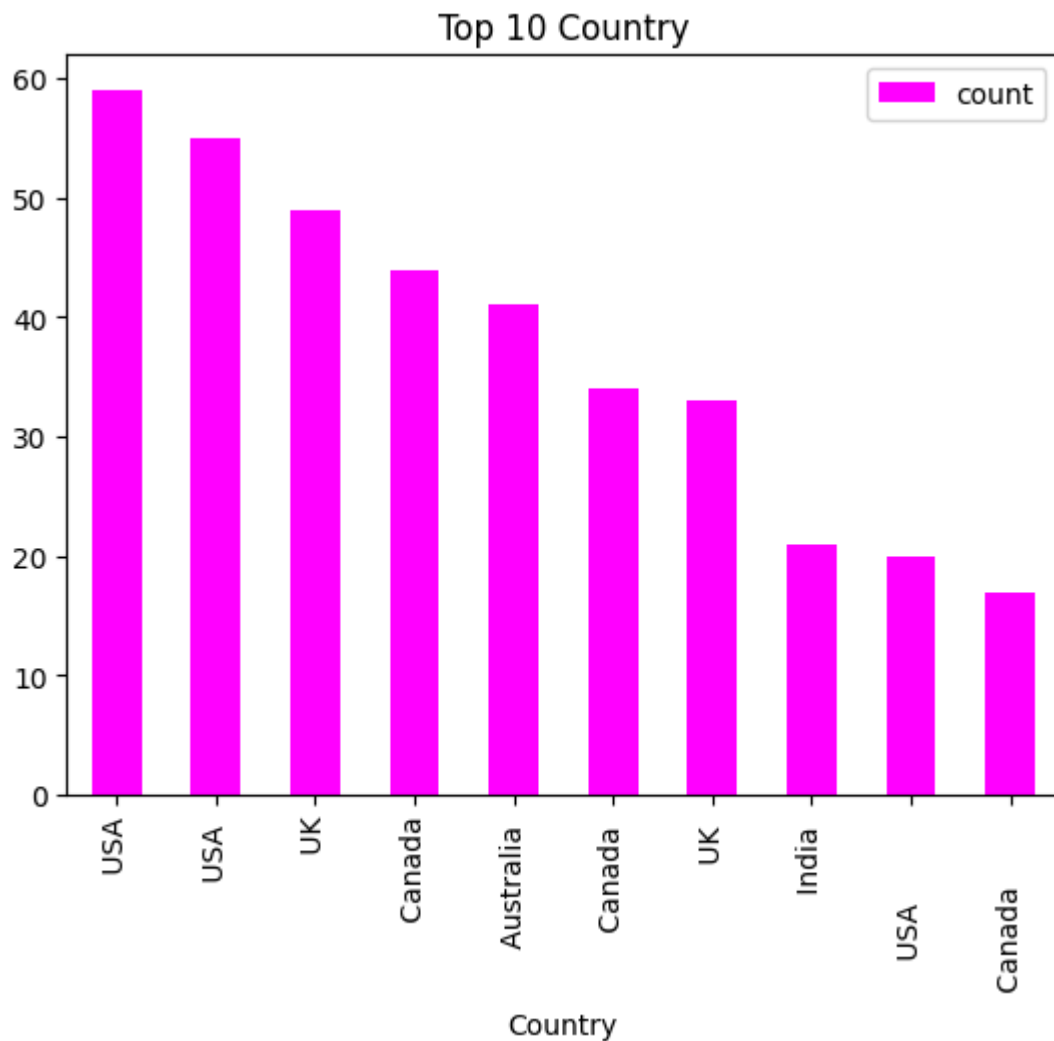
```
ax = sns.barplot(x = "Year" , y='Retweets', data = f,palette='gist_rainbow_r')
ax.bar_label(ax.containers[0])
plt.title("Retweets by Year" , fontsize =12 )
plt.show()
```



```
In [231... f=df.groupby('Year')['Likes'].sum().reset_index()
plt.figure(figsize = (8,5))
ax = sns.barplot(x = "Year" , y='Likes', data = f,palette='prism_r')
ax.bar_label(ax.containers[0])
plt.title("Likes by Year" , fontsize =12 )
plt.show()
```



```
In [78]: df['Country'].value_counts().nlargest(10).plot(kind='bar', cmap="spring")
plt.title('Top 10 Country')
plt.legend()
plt.show()
```



Data Manipulations

```
In [ ]:
```

```
In [17]: df['Timestamp'] = pd.to_datetime(df['Timestamp'])
df['Day'] = df['Timestamp'].dt.day
df['Month'] = df['Timestamp'].dt.month
df['Year'] = df['Timestamp'].dt.year
df['Hour'] = df['Hour'].astype('int64')
```

```
In [ ]:
```

```
In [18]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 732 entries, 0 to 731
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0.1          732 non-null   int64
1   Unnamed: 0            732 non-null   int64
2   Text                  732 non-null   object
3   Sentiment             732 non-null   object
4   Timestamp             732 non-null   datetime64[ns]
5   User                  732 non-null   object
6   Platform              732 non-null   object
7   Hashtags              732 non-null   object
8   Retweets              732 non-null   float64
9   Likes                 732 non-null   float64
10  Country               732 non-null   object
11  Year                  732 non-null   int32
12  Month                 732 non-null   int32
13  Day                   732 non-null   int32
14  Hour                  732 non-null   int64
dtypes: datetime64[ns](1), float64(2), int32(3), int64(3), object(6)
memory usage: 77.3+ KB

```

In [19]: `df.shape`

Out[19]: (732, 15)

In [20]: `df.columns`

Out[20]: Index(['Unnamed: 0.1', 'Unnamed: 0', 'Text', 'Sentiment', 'Timestamp', 'User', 'Platform', 'Hashtags', 'Retweets', 'Likes', 'Country', 'Year', 'Month', 'Day', 'Hour'], dtype='object')

In [21]: `df.drop(columns=["Unnamed: 0.1", "Unnamed: 0"], inplace=True)`

In [22]: `corr_matrix=df.corr(numeric_only=True)`
`corr_matrix`

Out[22]:

	Retweets	Likes	Year	Month	Day	Hour
Retweets	1.000000	0.998482	-0.039982	0.073265	0.009213	0.196955
Likes	0.998482	1.000000	-0.043415	0.066643	0.011489	0.195331
Year	-0.039982	-0.043415	1.000000	-0.314845	0.021973	-0.087470
Month	0.073265	0.066643	-0.314845	1.000000	-0.135873	0.137835
Day	0.009213	0.011489	0.021973	-0.135873	1.000000	0.044072
Hour	0.196955	0.195331	-0.087470	0.137835	0.044072	1.000000

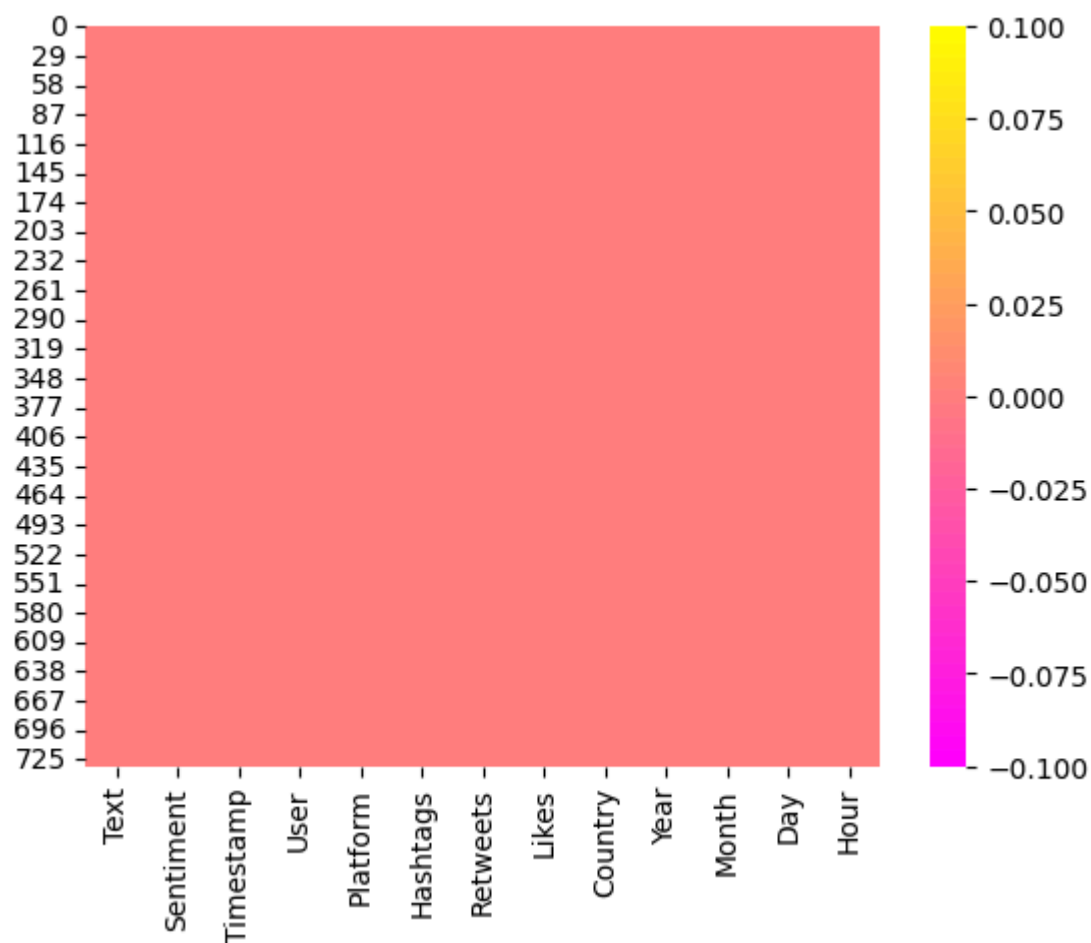
In [23]: `sns.heatmap(corr_matrix,annot=True)`

Out[23]: <Axes: >



In [226... `sns.heatmap(df.isnull(), cmap='spring')` *#Checked Null*

Out[226... `<Axes: >`



In [26]: `df.isnull().sum()`

```
Out[26]: Text      0
Sentiment  0
Timestamp  0
User       0
Platform   0
Hashtags   0
Retweets   0
Likes      0
Country    0
Year       0
Month      0
Day        0
Hour       0
dtype: int64
```

```
In [27]: df[['Text']]
```

```
Out[27]:
```

	Text
0	Enjoying a beautiful day at the park! ...
1	Traffic was terrible this morning. ...
2	Just finished an amazing workout! 🏋️ ...
3	Excited about the upcoming weekend getaway! ...
4	Trying out a new recipe for dinner tonight. ...
...	...
727	Collaborating on a science project that receiv...
728	Attending a surprise birthday party organized ...
729	Successfully fundraising for a school charity ...
730	Participating in a multicultural festival, cel...
731	Organizing a virtual talent show during challe...

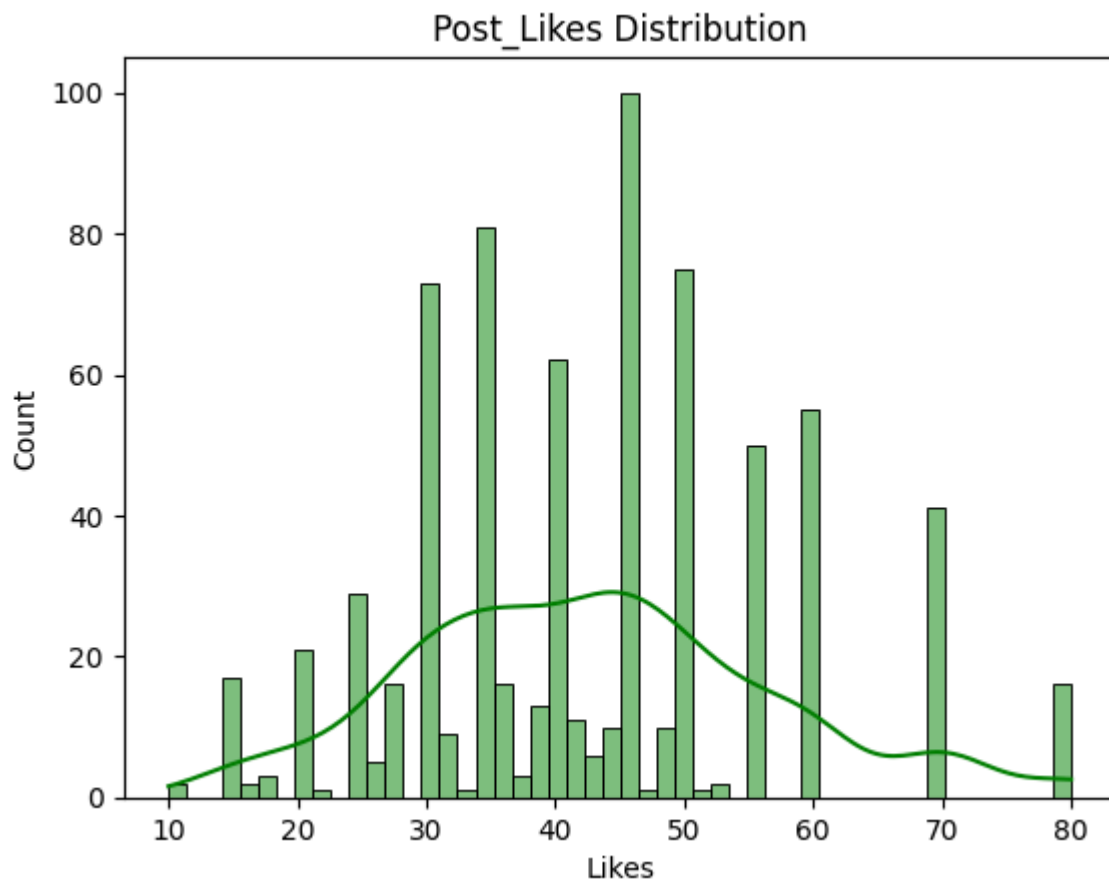
732 rows × 1 columns

```
In [28]: print(df['Sentiment'].value_counts())
```

```
Sentiment
Positive      44
Joy           42
Excitement    32
Happy         14
Neutral       14
..
Vibrancy      1
Culinary Adventure  1
Mesmerizing    1
Thrilling Journey  1
Winter Magic    1
Name: count, Length: 279, dtype: int64
```

```
In [84]: sns.histplot(df['Likes'],bins=50,color='green',kde=True)
plt.title("Post_Likes Distribution")
```

Out[84]: Text(0.5, 1.0, 'Post_Likes Distribution')



```
In [85]: p = df.groupby(['Platform', 'Year'])  
m=p.get_group(('Twitter', 2020))  
m
```

Out[85]:

	Text	Sentiment	Timestamp	User	Platform
166	Jealousy consumes me as I witness others' suc...	Jealousy	2020-08-12 11:30:00	EnviousObserver	Twitt
172	Helplessness sinks in as challenges pile up. ...	Helplessness	2020-06-20 14:20:00	OverwhelmedHeart	Twitt
234	Seeking serenity in the melody of raindrops, ...	Serenity	2020-04-12 19:30:00	RaindropHarmony	Twitt
237	Lost in the labyrinth of thoughts, confusion ...	Confusion	2020-06-05 12:15:00	ShadowedSeeker	Twitt
265	Proudly scaling the peaks of achievement, a m...	Proud	2020-01-05 08:45:00	PeakConqueror	Twitt
271	A free spirit soaring on the wings of dreams,...	Free-spirited	2020-06-10 10:05:00	DreamSoarer	Twitt
279	Proudly scaling the peaks of achievement, a m...	Proud	2020-01-05 08:45:00	PeakConqueror	Twitt
285	A free spirit soaring on the wings of dreams,...	Free-spirited	2020-06-10 10:05:00	DreamSoarer	Twitt
293	Proudly scaling the peaks of achievement, a m...	Proud	2020-01-05 08:45:00	PeakConqueror	Twitt
367	Happiness blooms like flowers in a garden on a...	Happiness	2020-02-10 11:15:00	SunnyDayEnthusiast	Twitt
411	Giddy with excitement as the first snowflakes ...	Excitement	2020-12-02 17:30:00	SnowfallEnthusiast	Twitt

	Text	Sentiment	Timestamp	User	Platfor
449	Avoiding the shards of shattered dreams, walki...	Resilience	2020-07-18 15:15:00	DreamTightrope	Twitt
486	Like a comet of inspiration, streaking through...	Inspiration	2020-05-28 16:00:00	CometCreativity	Twitt
514	Swaying to Ed Sheeran's acoustic melodies, a s...	Serenity	2020-08-18 19:00:00	SheeranMelodyLover	Twitt
539	As the closing scene unfolds, a sense of satis...	Satisfaction	2020-09-10 19:45:00	FilmCriticSatisfactionSeeker	Twitt
541	At the Olympics, the athlete's perseverance sh...	Inspiration	2020-08-08 19:45:00	OlympicsEnthusiastGoldMedalWinner	Twitt
564	In the world of science, a breakthrough discov...	Breakthrough	2020-05-20 16:30:00	ScienceEnthusiastDiscoveryMoment	Twitt

```
In [86]: likes_by_year = df.groupby('Year')['Likes'].sum().reset_index()
print(likes_by_year)
```

```

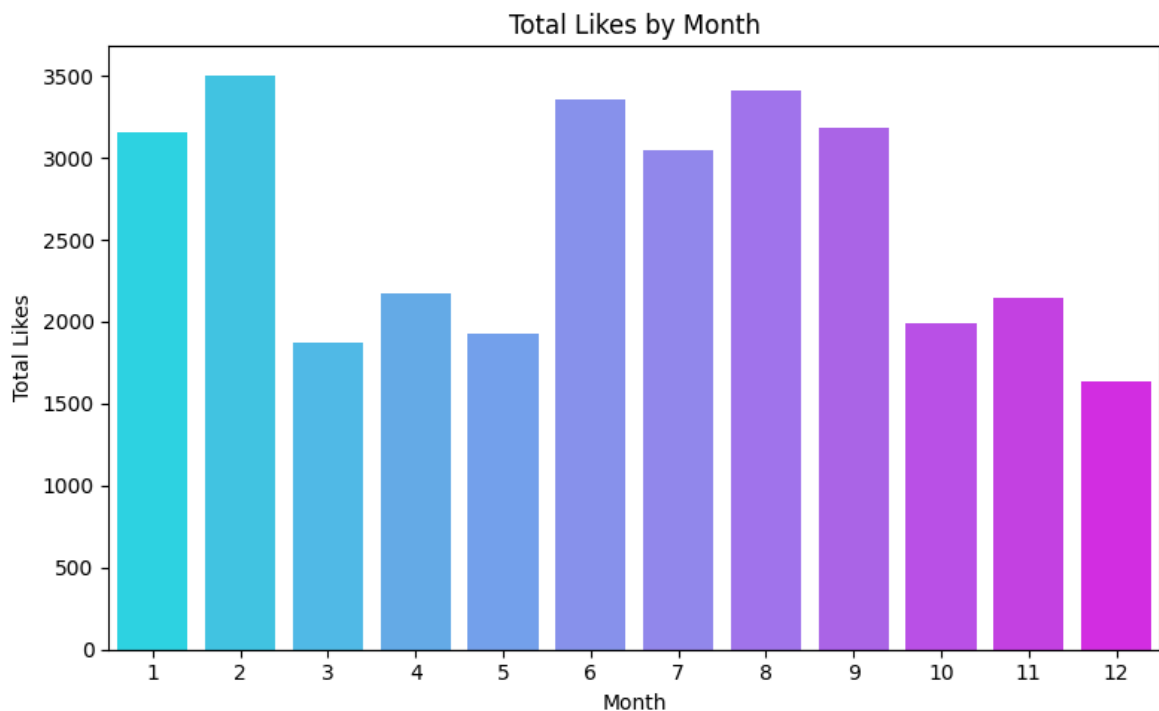
   Year  Likes
0  2010   160.0
1  2011   185.0
2  2012   160.0
3  2013   175.0
4  2014   160.0
5  2015   859.0
6  2016  1661.0
7  2017  1785.0
8  2018  2505.0
9  2019  3224.0
10 2020  2917.0
11 2021  2648.0
12 2022  2737.0
13 2023 12228.0
```

```
In [56]: retweets_by_year = df.groupby('Month')['Likes'].sum().reset_index()

plt.figure(figsize=(8, 5))
```

```
sns.barplot(data=retweets_by_year, x='Month', y='Likes',palette='cool')

plt.title("Total Likes by Month")
plt.xlabel("Month")
plt.ylabel("Total Likes")
plt.tight_layout()
plt.show()
```



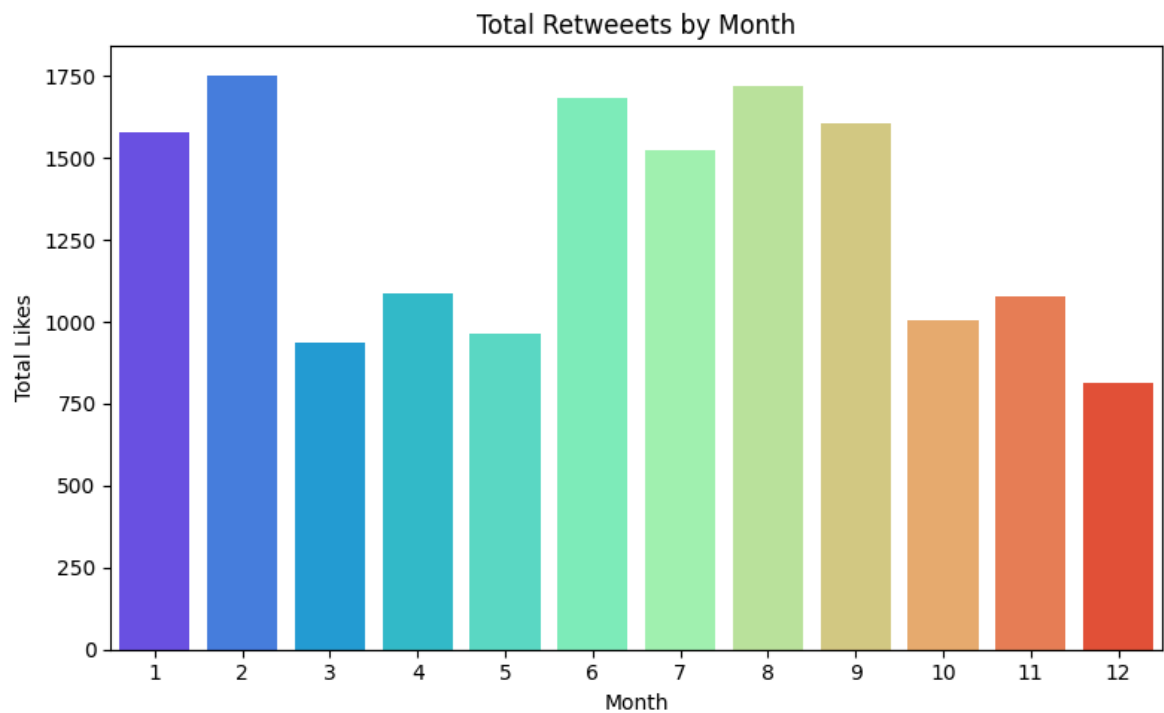
```
In [36]: likes_by_year = df.groupby('Month')['Retweets'].sum().reset_index()
print(likes_by_year)
```

	Month	Retweets
0	1	1577.0
1	2	1753.0
2	3	937.0
3	4	1087.0
4	5	963.0
5	6	1683.0
6	7	1523.0
7	8	1718.0
8	9	1606.0
9	10	1006.0
10	11	1076.0
11	12	815.0

```
In [53]: retweets_by_year = df.groupby('Month')['Retweets'].sum().reset_index()

plt.figure(figsize=(8, 5))
sns.barplot(data=retweets_by_year, x='Month', y='Retweets',palette='rainbow')

plt.title("Total Retweeets by Month")
plt.xlabel("Month")
plt.ylabel("Total Likes")
plt.tight_layout()
plt.show()
```



```
In [64]: engagement_by_year = df.groupby('Year')[['Likes', 'Retweets']].sum().reset_index
print(engagement_by_year)
```

	Year	Likes	Retweets
0	2010	160.0	80.0
1	2011	185.0	91.0
2	2012	160.0	80.0
3	2013	175.0	88.0
4	2014	160.0	80.0
5	2015	859.0	434.0
6	2016	1661.0	836.0
7	2017	1785.0	892.0
8	2018	2505.0	1253.0
9	2019	3224.0	1611.0
10	2020	2917.0	1458.0
11	2021	2648.0	1326.0
12	2022	2737.0	1367.0
13	2023	12228.0	6148.0

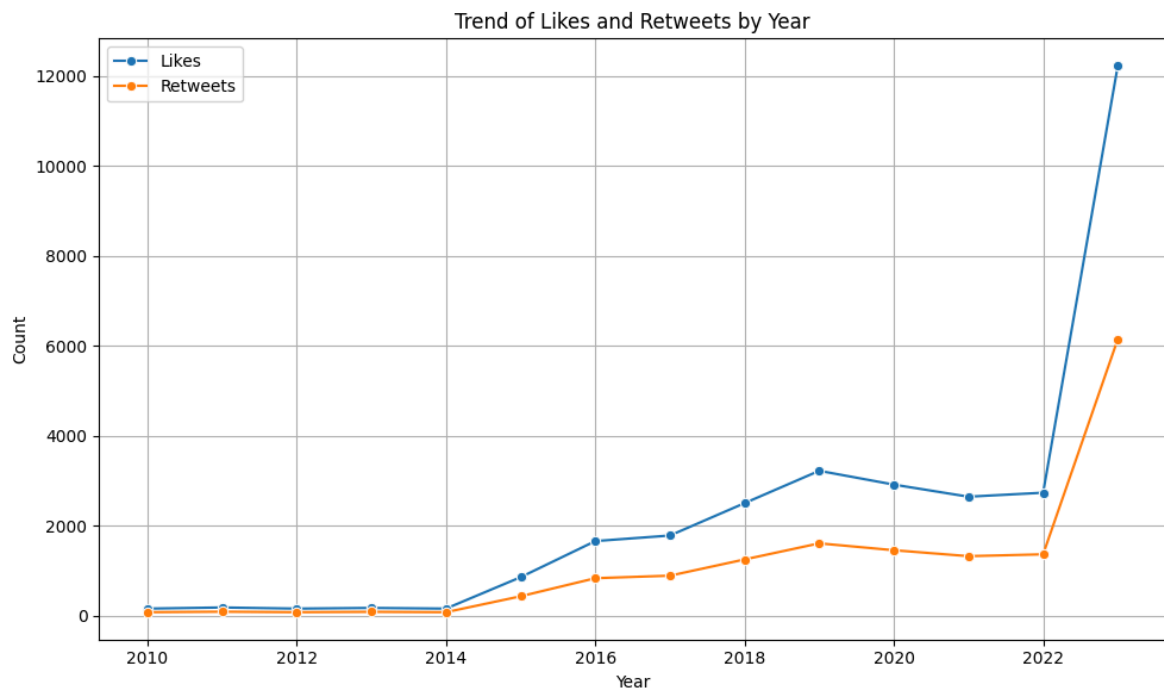
```
In [ ]:
```

```
In [74]: engagement_by_year = df.groupby('Year')[['Likes', 'Retweets']].sum().reset_index

plt.figure(figsize=(10, 6))

sns.lineplot(data=engagement_by_year, x='Year', y='Likes', marker='o', label='Li
sns.lineplot(data=engagement_by_year, x='Year', y='Retweets', marker='o', label=

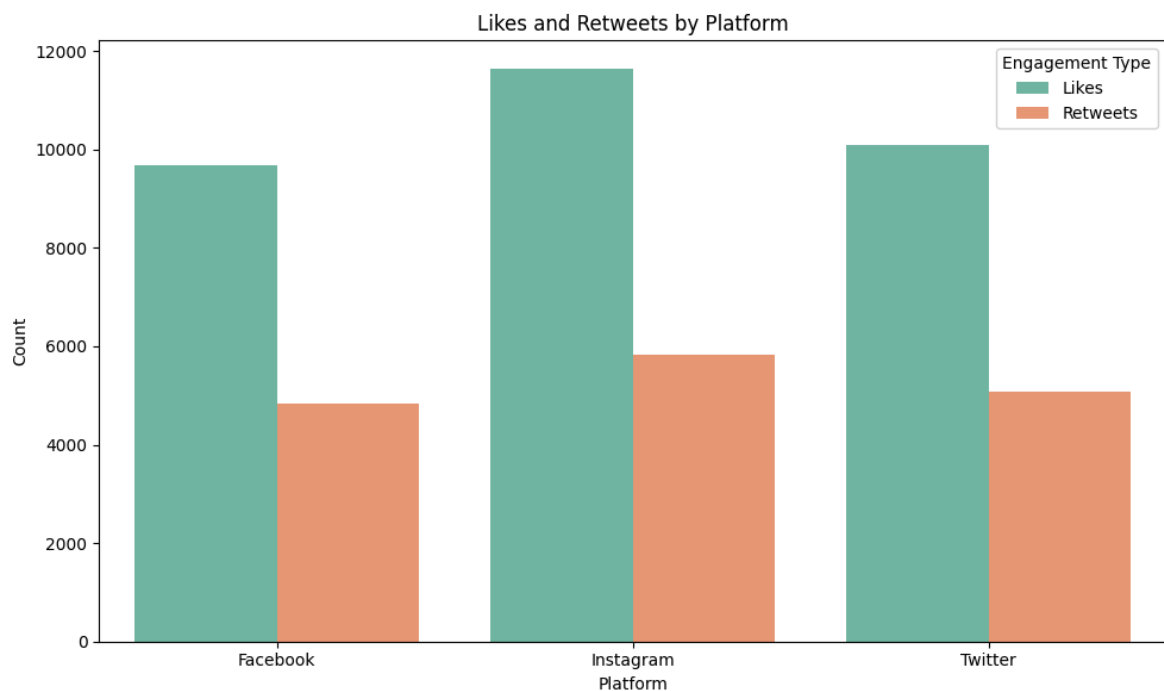
plt.title("Trend of Likes and Retweets by Year")
plt.xlabel("Year")
plt.ylabel("Count")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



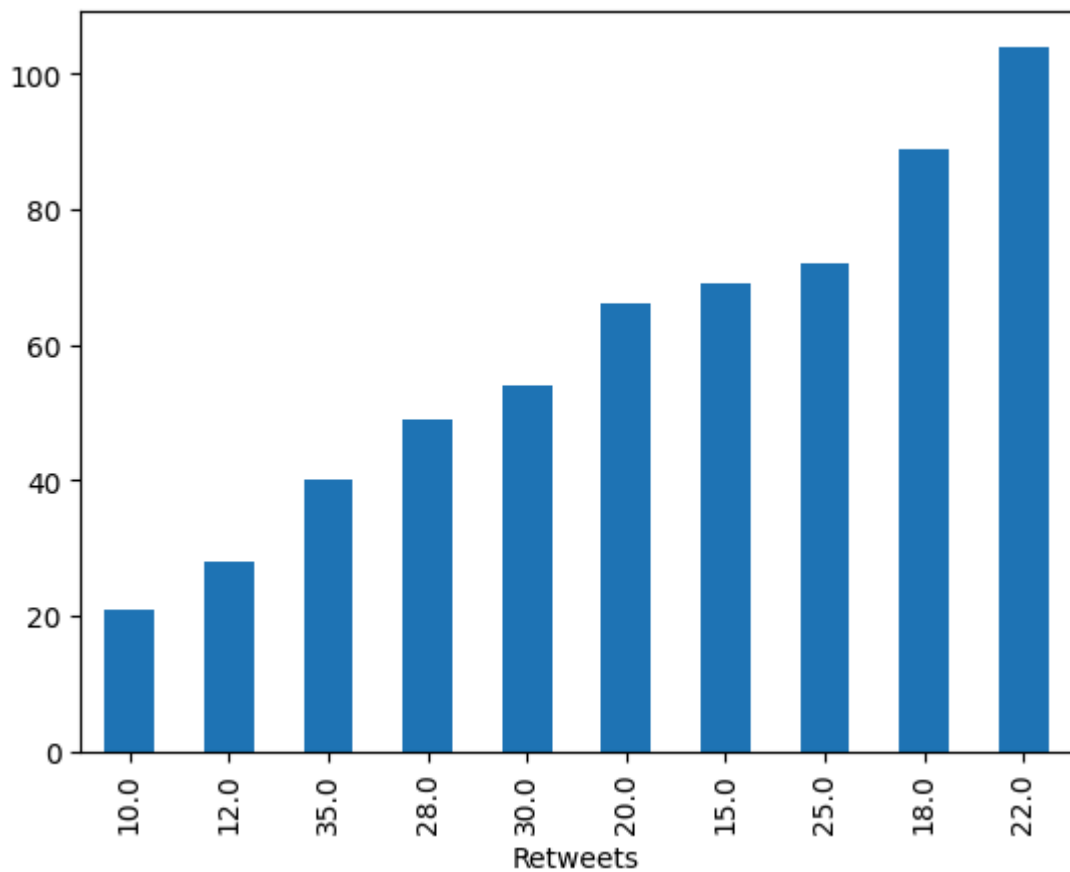
In []:

```
In [72]: engagement_by_year = df.groupby('Platform')[['Likes', 'Retweets']].sum().reset_index()
plt.figure(figsize=(10, 6))
bar_data = pd.melt(engagement_by_year, id_vars='Platform', value_vars=['Likes', 'Retweets'])
sns.barplot(data=bar_data, x='Platform', y='count', hue='type', palette='Set2')

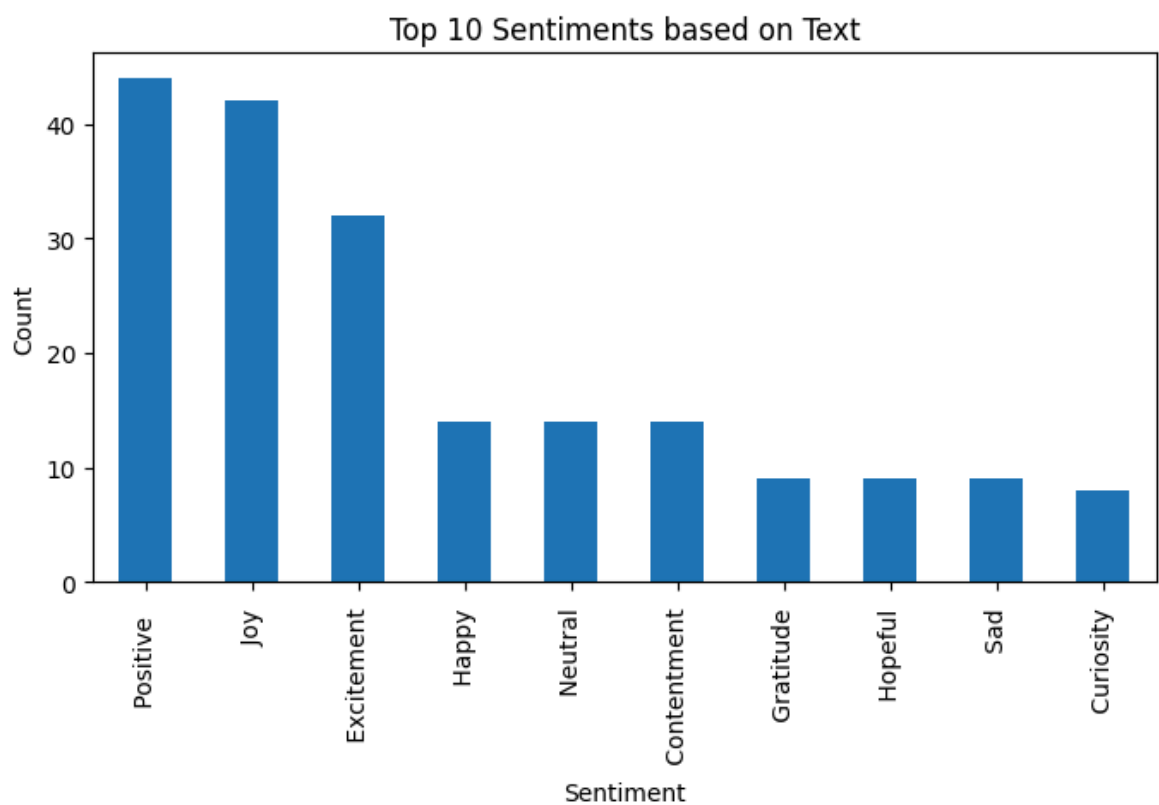
plt.title("Likes and Retweets by Platform")
plt.xlabel("Platform")
plt.ylabel("Count")
plt.legend(title="Engagement Type")
plt.tight_layout()
plt.show()
```



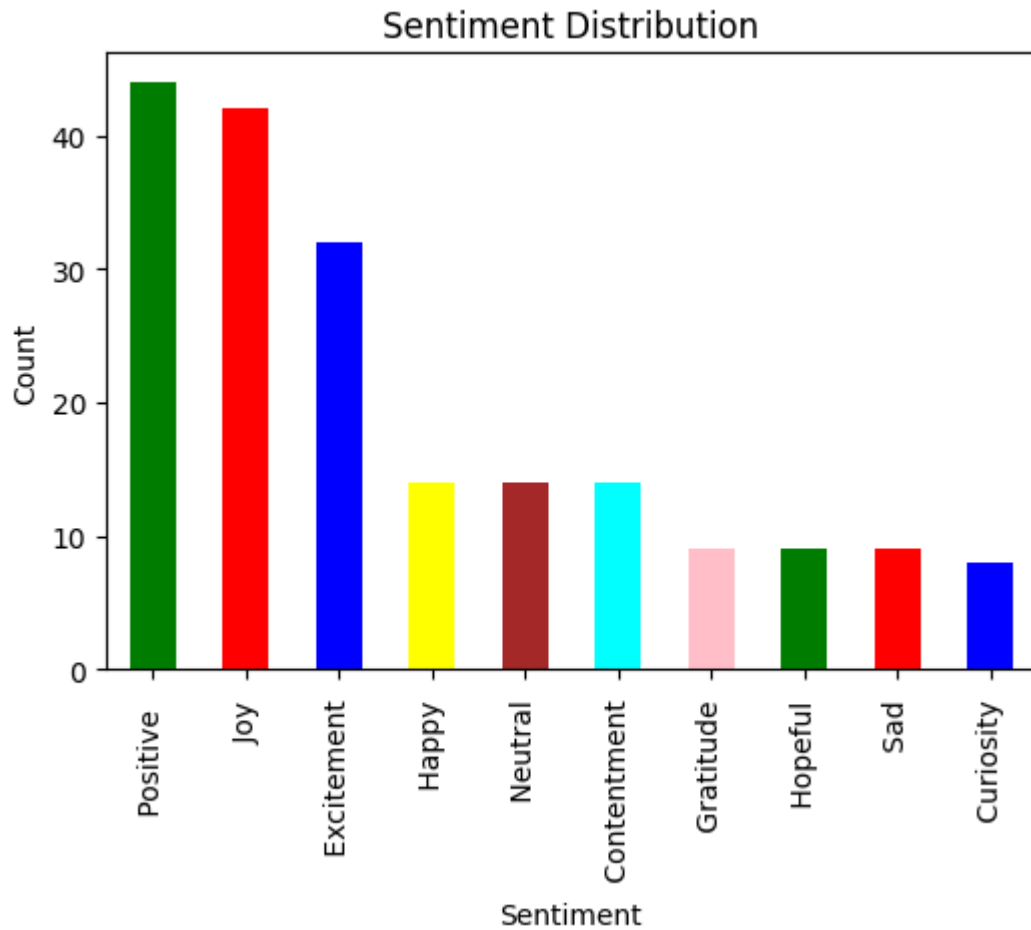
```
In [92]: H_retweet = df.groupby('Hashtags')['Retweets'].max().value_counts().nlargest(10)
```



```
In [110... plt.figure(figsize=(8,4))
df['Sentiment'].value_counts().nlargest(10).plot(kind='bar')
plt.title('Top 10 Sentiments based on Text')
plt.xlabel('Sentiment')
plt.ylabel('Count')
plt.show()
```



```
In [99]: # Visualization - Sentiment Distribution
plt.figure(figsize=(6,4))
df['Sentiment'].value_counts().nlargest(10).plot(kind='bar', color=['green', 'red', 'blue', 'yellow', 'brown', 'cyan', 'pink', 'darkgreen', 'darkred', 'darkblue'])
plt.title("Sentiment Distribution")
plt.xlabel("Sentiment")
plt.ylabel("Count")
plt.show()
```



```
In [100]: df.head(5)
```

Out[100...

	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes
0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30
1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10
2	Just finished an amazing workout! 🏋️ ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40
3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	8.0	15
4	Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	12.0	25



In [101...

```
## Investigate trends over time using the "Timestamp" column.      ## eliminate S
df['Time'] = df['Timestamp'].dt.strftime("%H:%M")
df['Date'] = df['Timestamp'].dt.strftime("%Y-%m-%d")
```

In [102...

```
df.head(20)
```

Out[102...

	Text	Sentiment	Timestamp	User	Platform	Hashtags	Ret
0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	
1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	
2	Just finished an amazing workout! 🍌 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	
3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	
4	Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	
5	Feeling grateful for the little things in lif...	Positive	2023-01-16 09:10:00	GratitudeNow	Twitter	#Gratitude #PositiveVibes	
6	Rainy days call for cozy blankets and hot coc...	Positive	2023-01-16 14:45:00	RainyDays	Facebook	#RainyDays #Cozy	
7	The new movie release is a must-watch! ...	Positive	2023-01-16 19:30:00	MovieBuff	Instagram	#MovieNight #MustWatch	
8	Political discussions heating up on the timel...	Negative	2023-01-17 08:00:00	DebateTalk	Twitter	#Politics #Debate	
9	Missing summer vibes and beach days. ...	Neutral	2023-01-17 12:20:00	BeachLover	Facebook	#Summer #BeachDays	
10	Just published	Positive	2023-01-17 15:15:00	BloggerX	Instagram	#Blogging #NewPost	

	Text	Sentiment	Timestamp	User	Platform	Hashtags	Ret
	a new blog post. Check it out!...						
11	Feeling a bit under the weather today. ...	Negative	2023-01-18 10:30:00	WellnessCheck	Twitter	#SickDay #Health	
12	Exploring the city's hidden gems. ...	Positive	2023-01-18 14:50:00	UrbanExplorer	Facebook	#CityExplore #HiddenGems	
13	New year, new fitness goals! 🏋️ ...	Positive	2023-01-18 18:00:00	FitJourney	Instagram	#NewYear #FitnessGoals	
14	Technology is changing the way we live. ...	Neutral	2023-01-19 09:45:00	TechEnthusiast	Twitter	#Tech #Innovation	
15	Reflecting on the past and looking ahead. ...	Positive	2023-01-19 13:20:00	Reflections	Facebook	#Reflection #Future	
16	Just adopted a cute furry friend! 🐾 ...	Positive	2023-01-19 17:10:00	PetAdopter	Instagram	#PetAdoption #FurryFriend	
17	Late-night gaming session with friends. ...	Positive	2023-01-20 00:05:00	GamerX	Twitter	#Gaming #LateNight	
18	Attending a virtual conference on AI. ...	Neutral	2023-01-20 11:30:00	TechConference	Facebook	#AI #TechConference	
19	Winter blues got me feeling low. ...	Negative	2023-01-20 15:15:00	WinterBlues	Instagram	#WinterBlues #Mood	

```
In [103... df.columns
```

```
Out[103... Index(['Text', 'Sentiment', 'Timestamp', 'User', 'Platform', 'Hashtags',
      'Retweets', 'Likes', 'Country', 'Year', 'Month', 'Day', 'Hour', 'Time',
      'Date'],
      dtype='object')
```

```
In [104... df.drop(columns="Time",inplace=True)
```

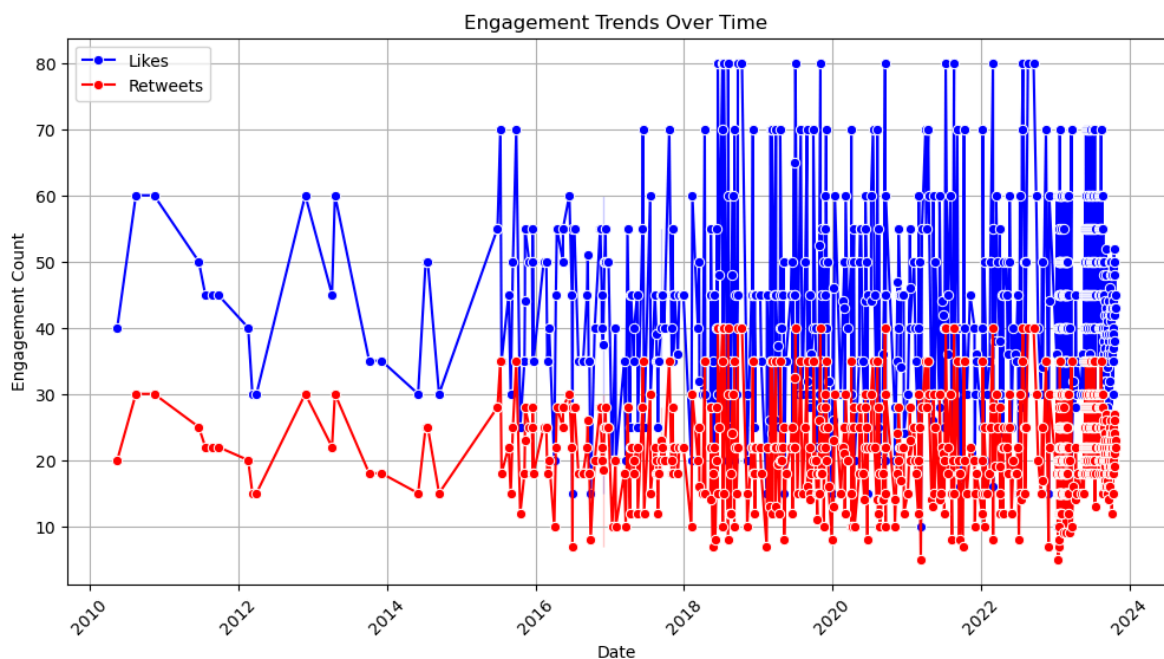
By User Behaviour

```
In [344... plt.figure(figsize=(12,6))

# Plot Likes over time
sns.lineplot(x=df["Timestamp"], y=df["Likes"], label="Likes", marker="o", color=

# Plot Retweets over time
sns.lineplot(x=df["Timestamp"], y=df["Retweets"], label="Retweets", marker="o",

plt.xticks(rotation=45)
plt.xlabel("Date")
plt.ylabel("Engagement Count")
plt.title("Engagement Trends Over Time")
plt.legend()
plt.grid(True)
plt.show()
```



Most Liked & Retweets

```
In [353... # Top 5 posts with most Likes
top_liked = [df.nlargest(5, "Likes")]["Timestamp", "Text", "Likes", "Retweets"]

# Top 5 posts with most retweets
top_retweeted = [df.nlargest(5, "Retweets")]["Timestamp", "Text", "Likes", "Retw

print("***Top 5 Most Liked Posts:***")
print(top_liked)
```

```
print("\n**Top 5 Most Retweeted Posts:**")
print(top_retweeted)
```

****Top 5 Most Liked Posts:****

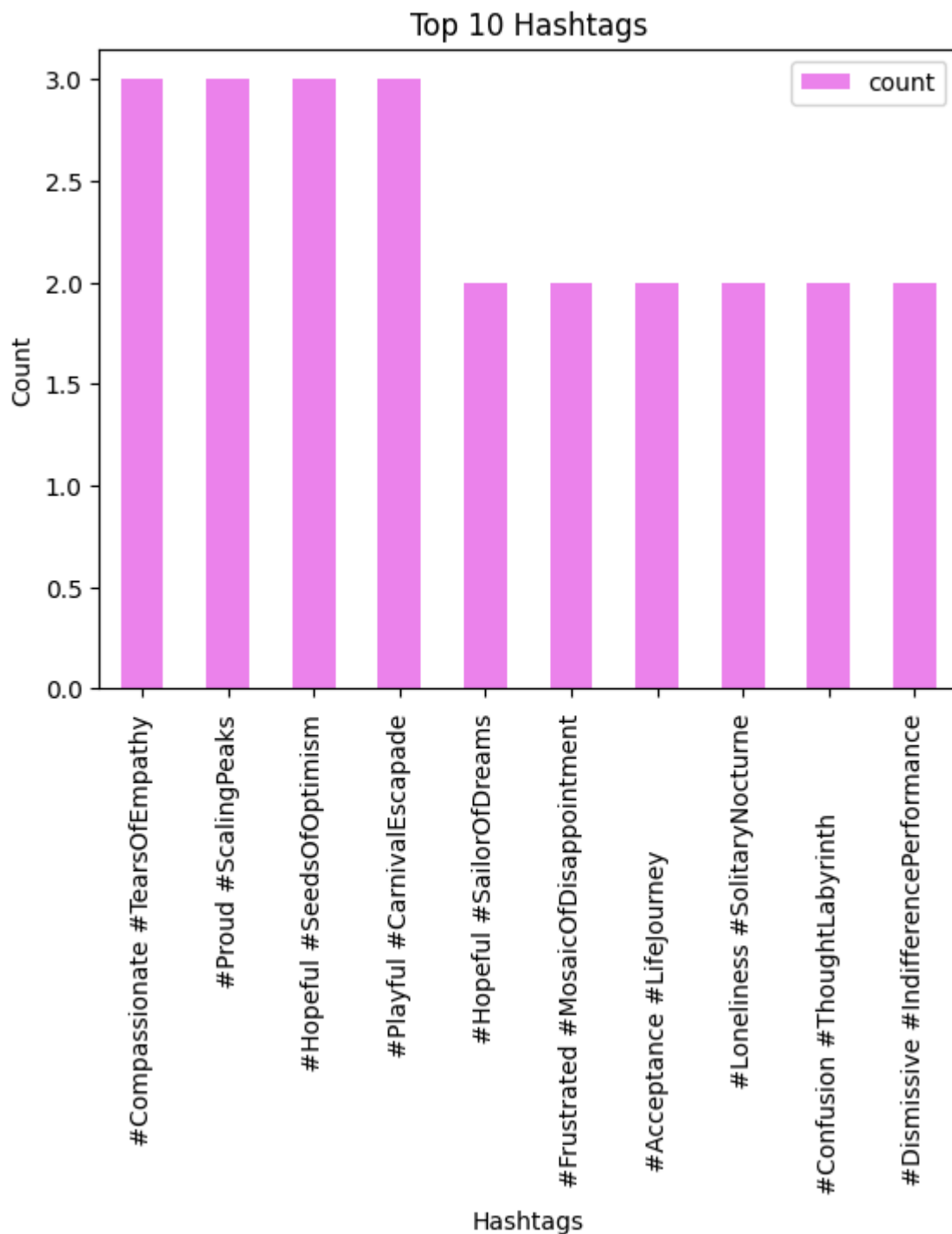
	Timestamp	Text \	Likes	Retweets
335	2020-09-15 14:45:00	Thrilled to witness the grandeur of a cultural...	80.0	40.0
345	2022-02-28 07:15:00	Motivated to achieve fitness goals after an in...	80.0	40.0
355	2022-07-25 10:00:00	Anticipation for an upcoming adventure in an e...	80.0	40.0
368	2018-09-22 16:30:00	Elation over discovering a rare book in a quai...	80.0	40.0
382	2018-07-08 23:00:00	A sense of wonder at the vastness of the cosmo...	80.0	40.0

****Top 5 Most Retweeted Posts:****

	Timestamp	Text \	Likes	Retweets
335	2020-09-15 14:45:00	Thrilled to witness the grandeur of a cultural...	80.0	40.0
345	2022-02-28 07:15:00	Motivated to achieve fitness goals after an in...	80.0	40.0
355	2022-07-25 10:00:00	Anticipation for an upcoming adventure in an e...	80.0	40.0
368	2018-09-22 16:30:00	Elation over discovering a rare book in a quai...	80.0	40.0
382	2018-07-08 23:00:00	A sense of wonder at the vastness of the cosmo...	80.0	40.0

In [355... `import seaborn as sns`

In [183... `df['Hashtags'].value_counts().nlargest(10).plot(kind='bar', color="violet")`
`plt.title('Top 10 Hashtags')`
`plt.xlabel('Hashtags')`
`plt.ylabel('Count')`
`plt.legend()`
`plt.show()`

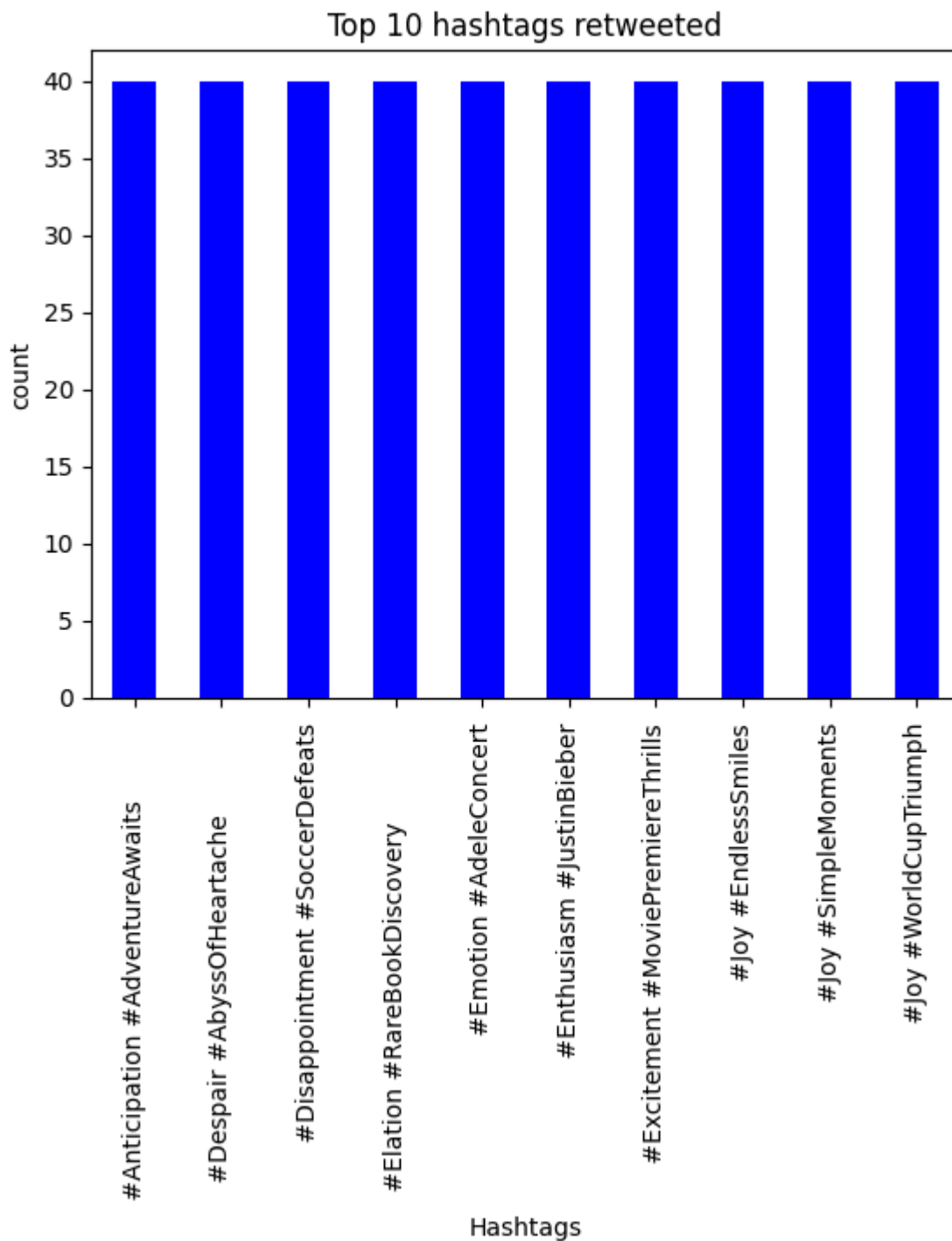


```
In [122...] numerical_columns = df[['Day', 'Month', 'Year', 'Likes', 'Retweets']]

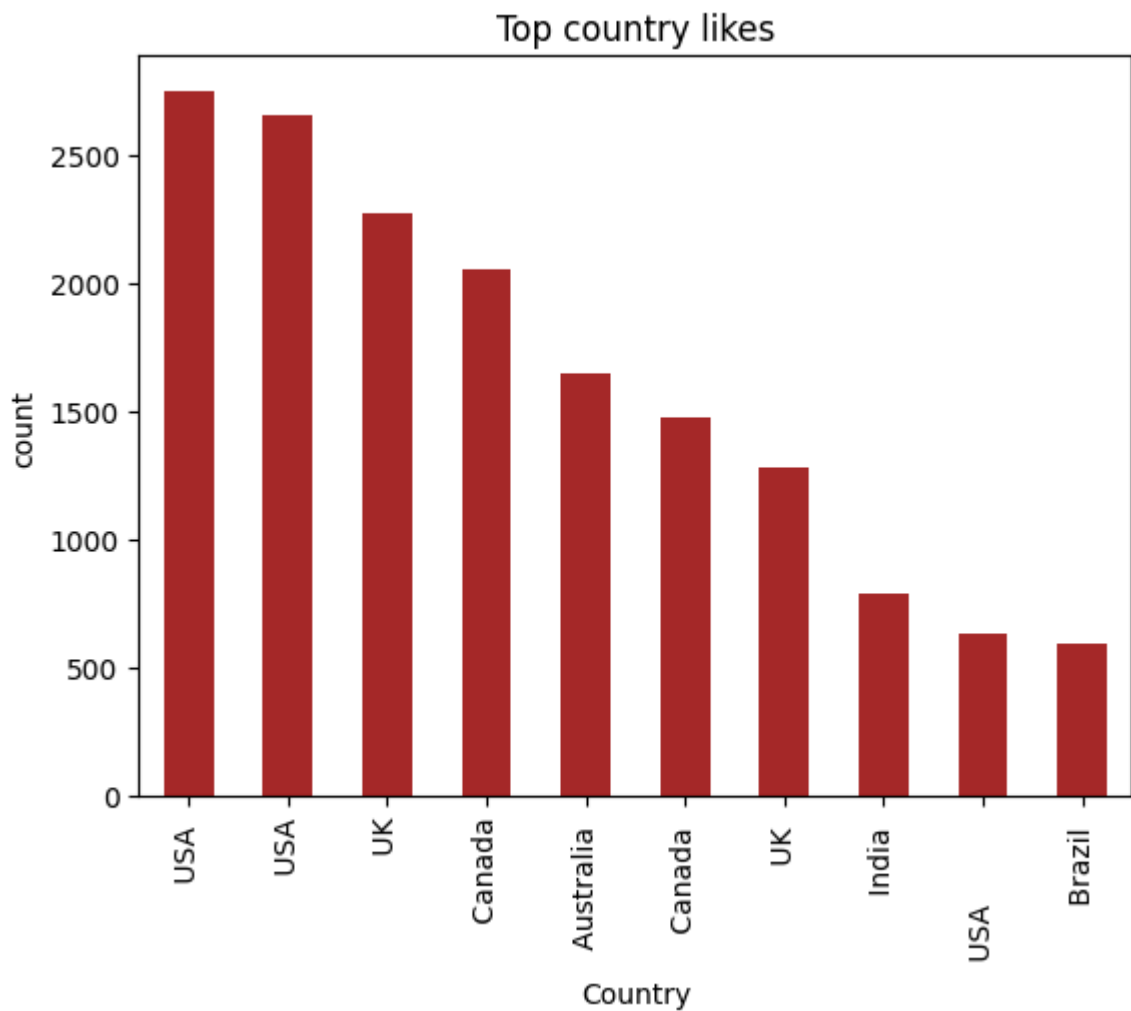
for col in numerical_columns.columns:
    print(f"Minimum {col}: {df[col].min()} | Maximum {col}: {df[col].max()}")
```

```
Minimum Day: 1 | Maximum Day: 31
Minimum Month: 1 | Maximum Month: 12
Minimum Year: 2010 | Maximum Year: 2023
Minimum Likes: 10.0 | Maximum Likes: 80.0
Minimum Retweets: 5.0 | Maximum Retweets: 40.0
```

```
In [179...] H_R=df.groupby('Hashtags')['Retweets'].max().nlargest(10).sort_values(ascending=
plt.title('Top 10 hashtags retweeted')
plt.xlabel('Hashtags')
plt.ylabel('count')
plt.show()
```

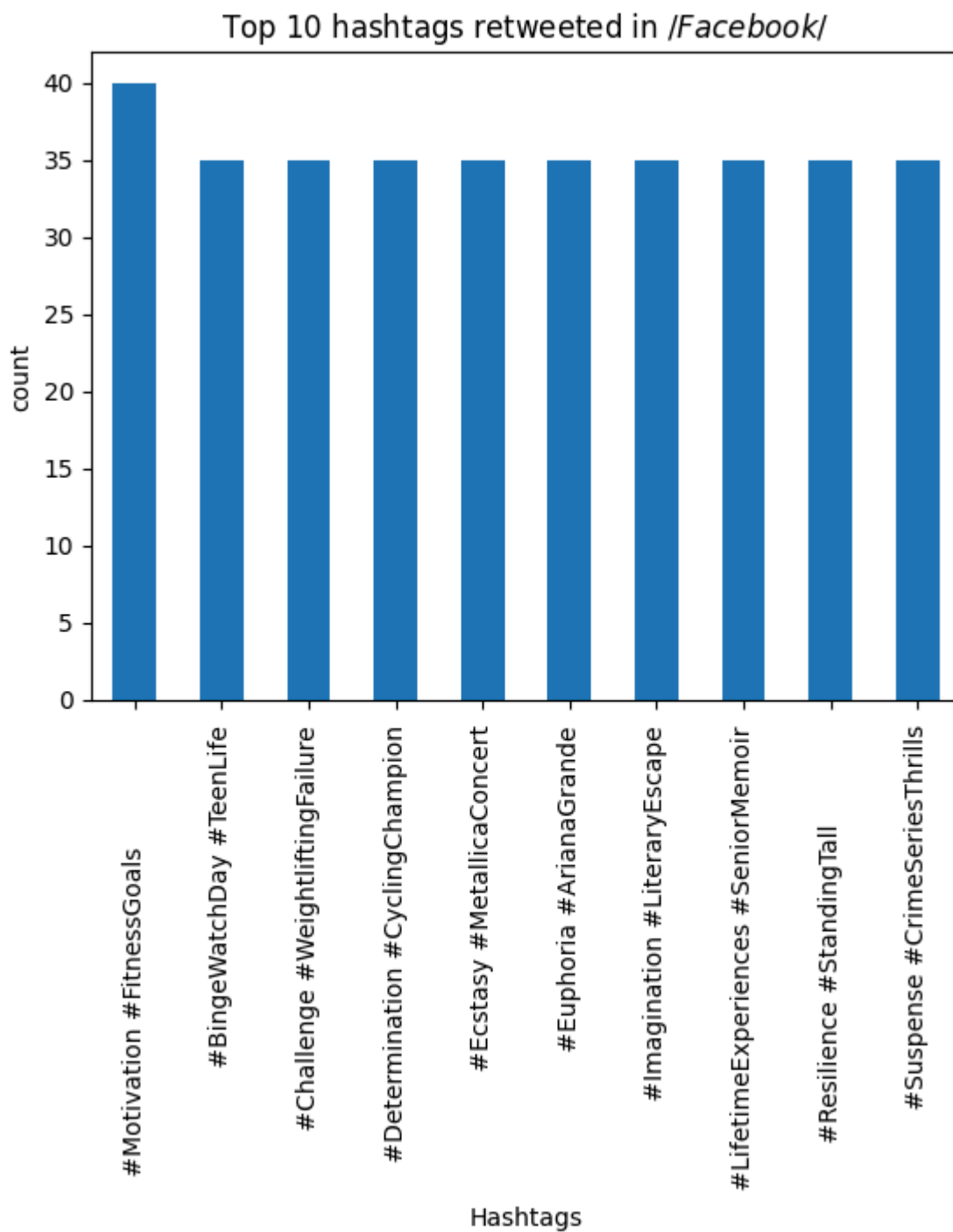


```
In [178... top_country_likes=df.groupby('Country')['Likes'].sum().nlargest(10)
top_country_likes.plot(kind='bar',color='brown')
plt.title('Top country likes')
plt.xlabel('Country')
plt.ylabel('count')
plt.show()
```

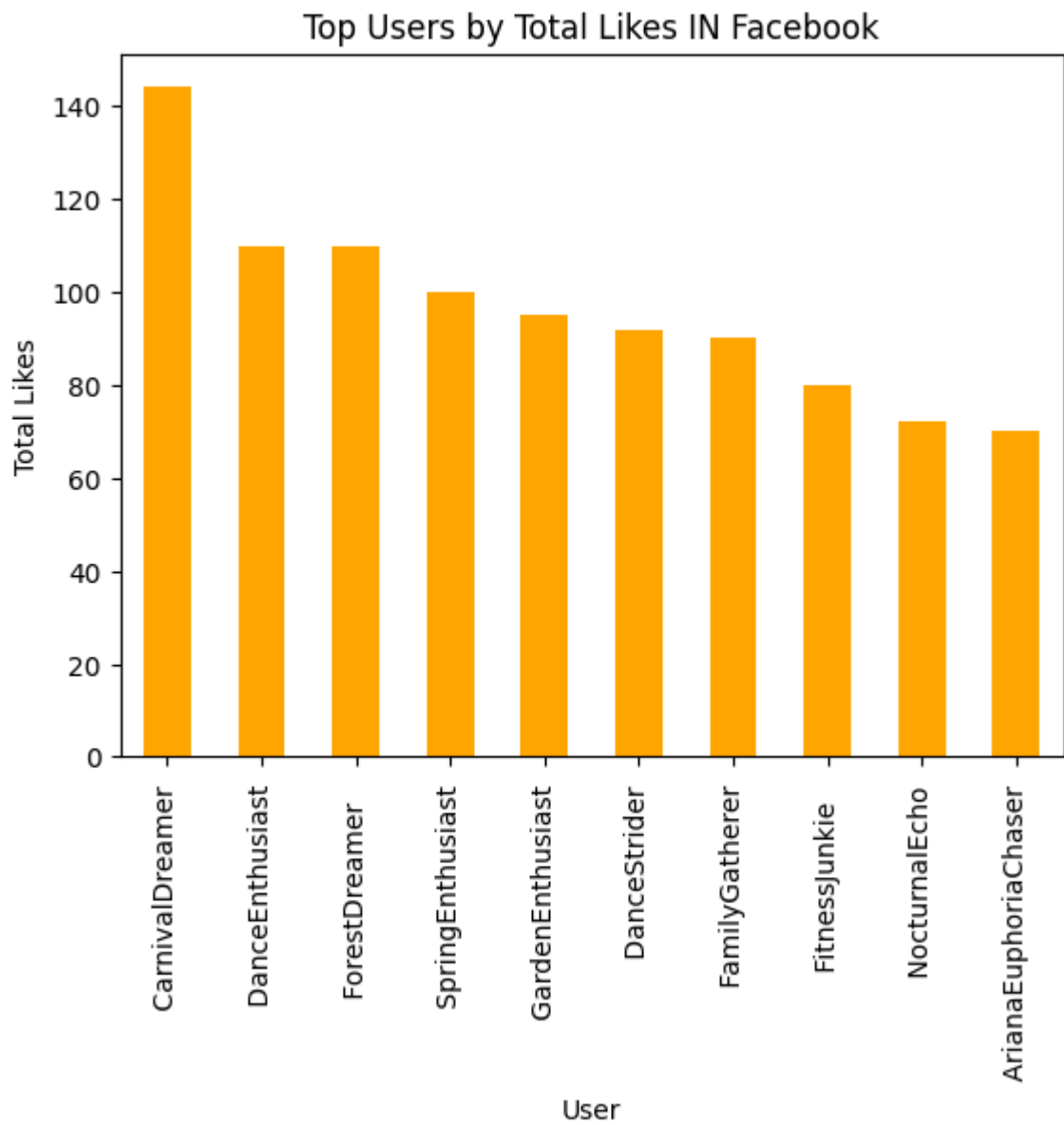


```
In [174... Facebook=df[df['Platform']=='Facebook']
Twitter=df[df['Platform']=='Twitter']
Instagram=df[df['Platform']=='Instagram']
```

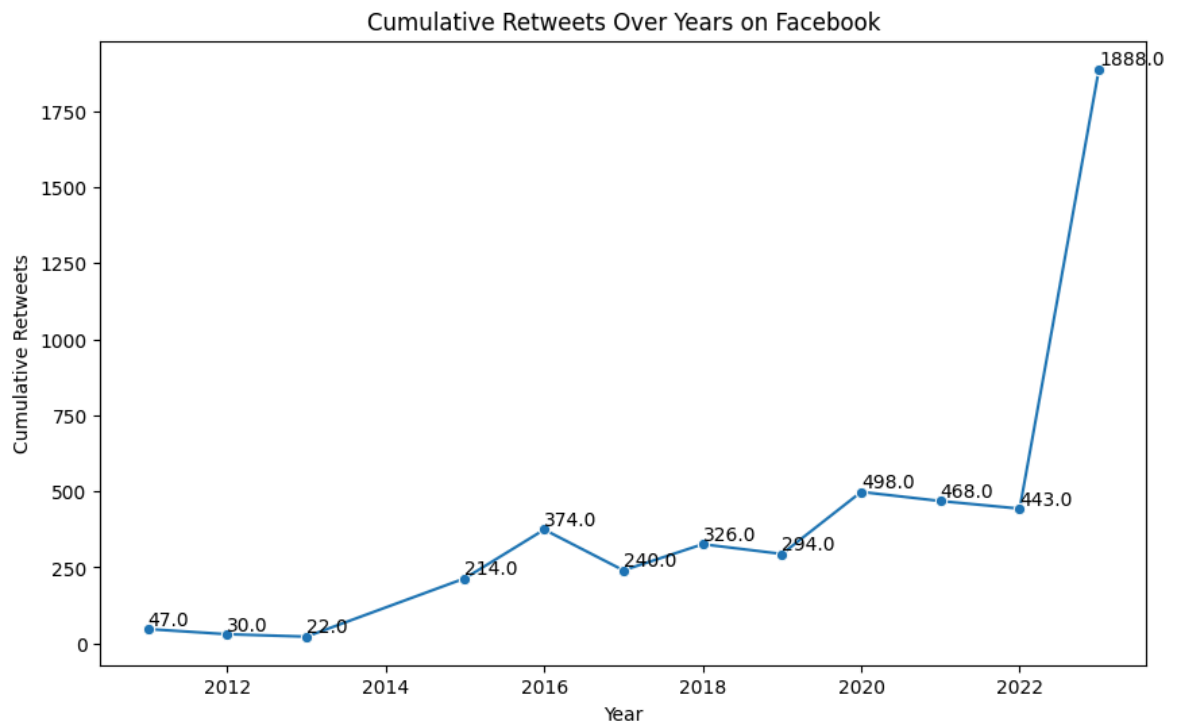
```
In [175... H_R_f=Facebook.groupby('Hashtags')['Retweets'].max().nlargest(10).sort_values(as
H_R_f.plot(kind='bar')
plt.title('Top 10 hashtags retweeted in $/ Facebook $/')
plt.xlabel('Hashtags')
plt.ylabel('count')
plt.show()
```



```
In [176... top_likes_platform_F = Facebook.groupby('User')['Likes'].sum().nlargest(10)
top_likes_platform_F.plot(kind='bar', color="orange")
plt.title('Top Users by Total Likes IN Facebook')
plt.xlabel('User')
plt.ylabel('Total Likes')
plt.show()
```



In [129...

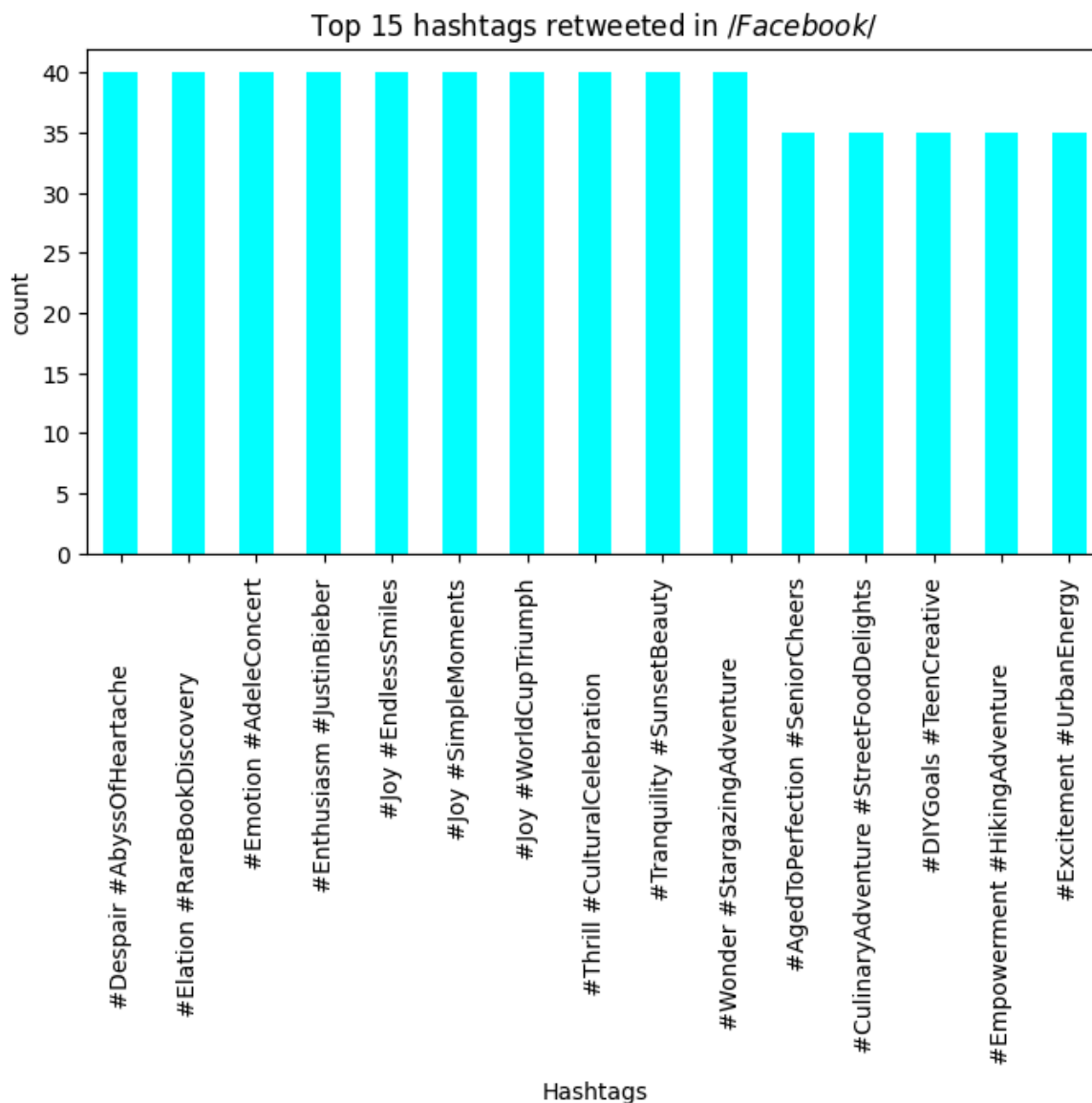


In [142...

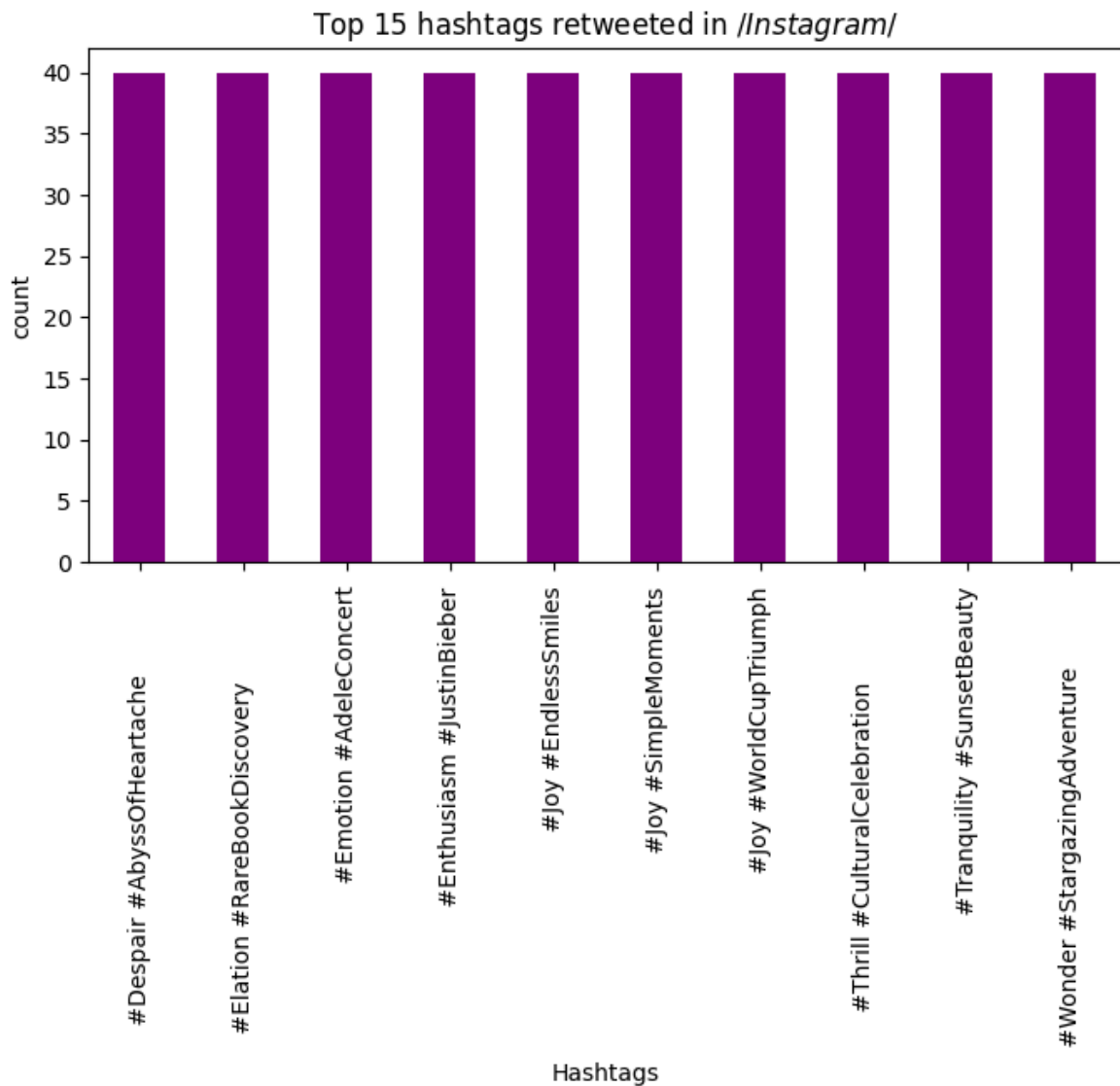
```

facebook= df[df['Platform'] == 'Facebook']
H_R_i=instagram.groupby('Hashtags')['Retweets'].max().nlargest(15).sort_values(a
plt.figure(figsize=(8,4))
H_R_i.plot(kind='bar',color="cyan")
plt.title('Top 15 hashtags retweeted in $/ Facebook $/')
plt.xlabel('Hashtags')
plt.ylabel('count')
plt.show()

```



```
In [143...
instagram= df[df['Platform'] == 'Instagram']
H_R_i=instagram.groupby('Hashtags')['Retweets'].max().nlargest(10).sort_values(a
plt.figure(figsize=(8,4))
H_R_i.plot(kind='bar',color="purple")
plt.title('Top 15 hashtags retweeted in $/ Instagram $/')
plt.xlabel('Hashtags')
plt.ylabel('count')
plt.show()
```

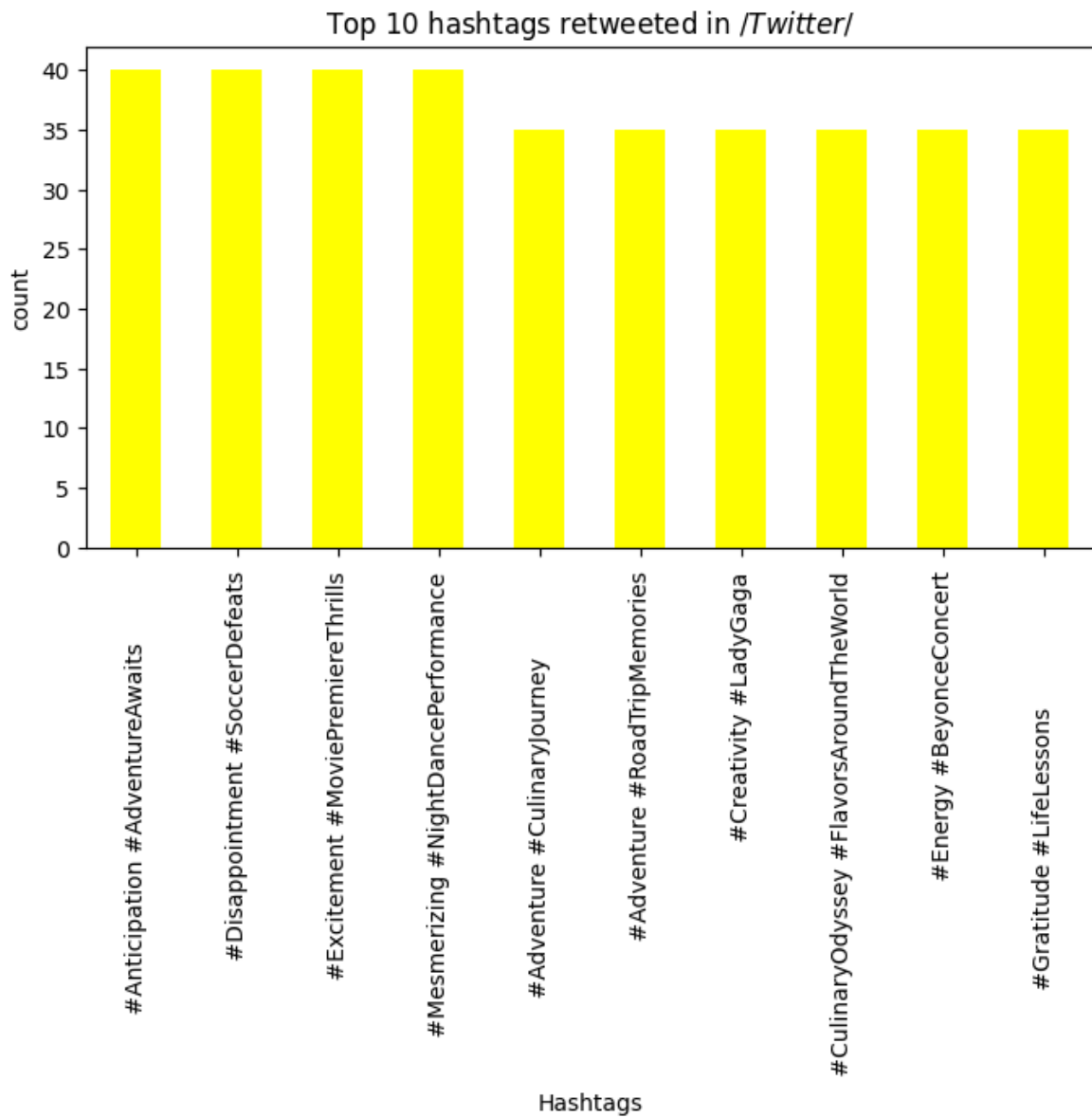


In [144...

```

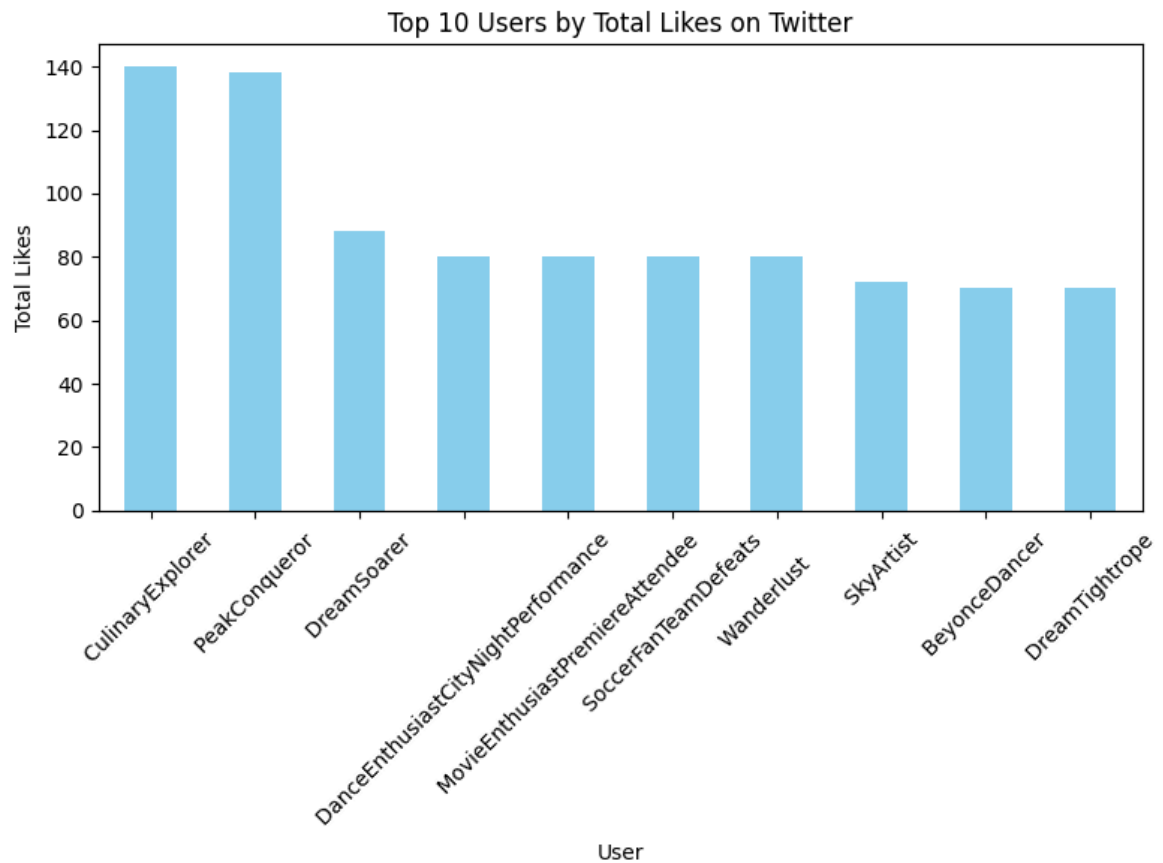
twitter= df[df['Platform'] == 'Twitter']
H_R_t=twitter.groupby('Hashtags')['Retweets'].max().nlargest(10).sort_values(asc
plt.figure(figsize=(8,4))
H_R_t.plot(kind='bar',color="yellow")
plt.title('Top 10 hashtags retweeted in $/ Twitter $/')
plt.xlabel('Hashtags')
plt.ylabel('count')
plt.show()

```



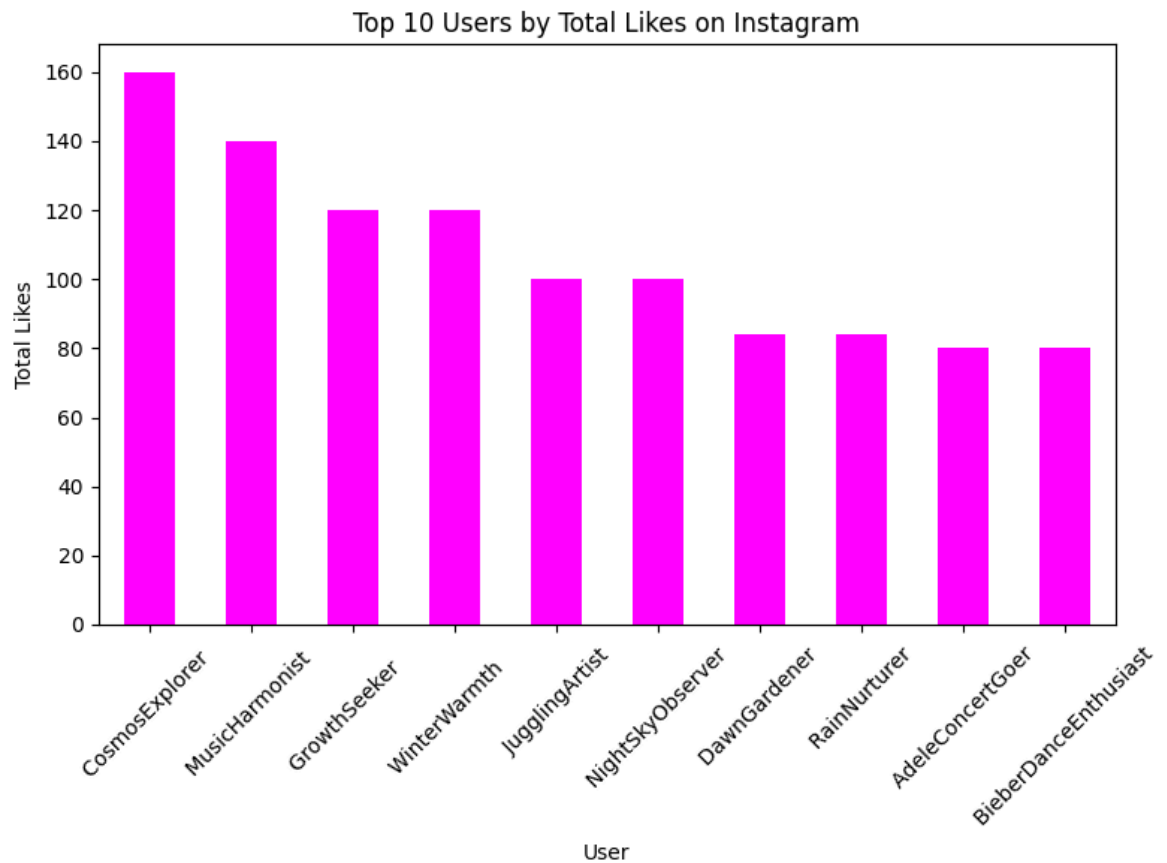
```
In [146... twitter_data = df[df['Platform'] == 'Twitter']
top_likes_platform_t = twitter_data.groupby('User')['Likes'].sum().nlargest(10)

plt.figure(figsize=(8,6))
top_likes_platform_t.plot(kind='bar', color='skyblue')
plt.title('Top 10 Users by Total Likes on Twitter')
plt.xlabel('User')
plt.ylabel('Total Likes')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [147...
instagram_data = df[df['Platform'] == 'Instagram']
top_likes_platform_t = instagram_data.groupby('User')['Likes'].sum().nlargest(10)

plt.figure(figsize=(8,6))
top_likes_platform_t.plot(kind='bar', color='magenta')
plt.title('Top 10 Users by Total Likes on Instagram')
plt.xlabel('User')
plt.ylabel('Total Likes')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In []:

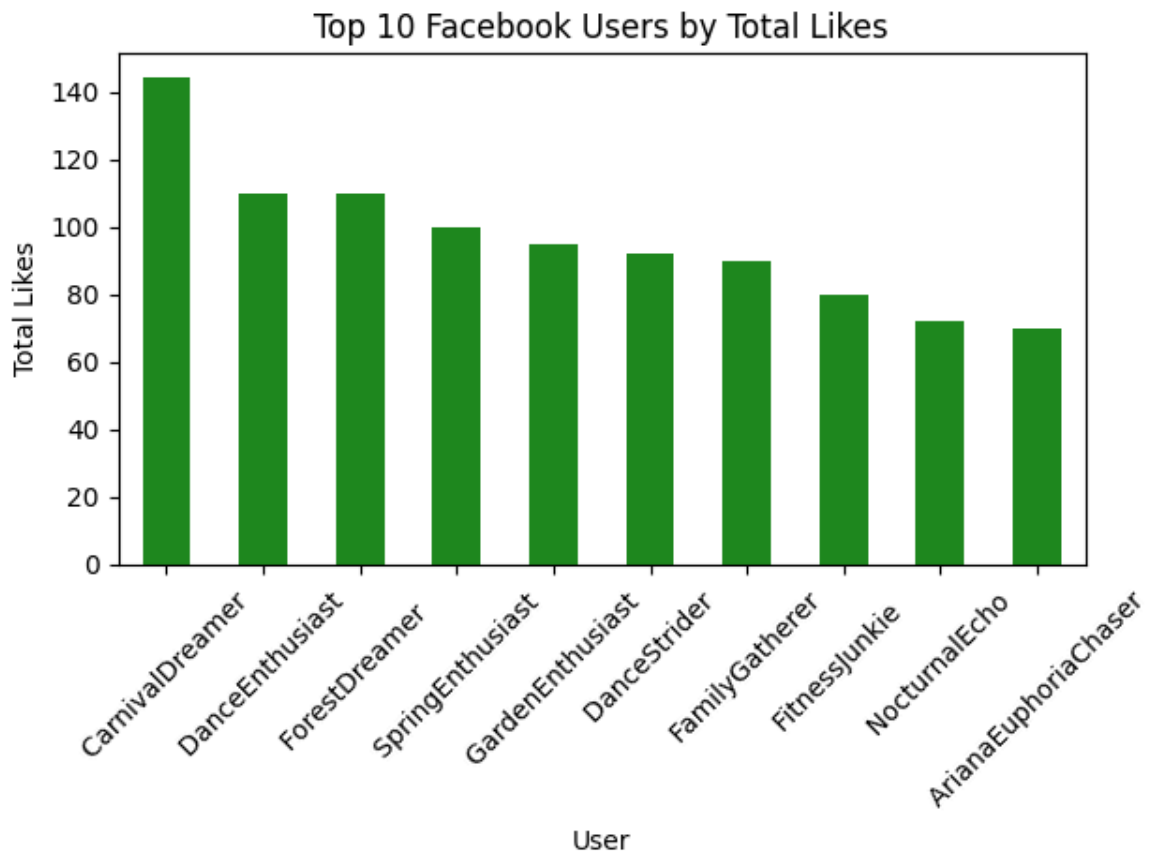
In [192...

```
import matplotlib.pyplot as plt

# Filter data for Twitter platform
facebook_data = df[df['Platform'] == 'Facebook']

# Group by User and sum Likes
top_likes_platform_t = facebook_data.groupby('User')['Likes'].sum().nlargest(10)

# Plotting
top_likes_platform_t.plot(kind='bar', color='forestgreen')
plt.title('Top 10 Facebook Users by Total Likes')
plt.xlabel('User')
plt.ylabel('Total Likes')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

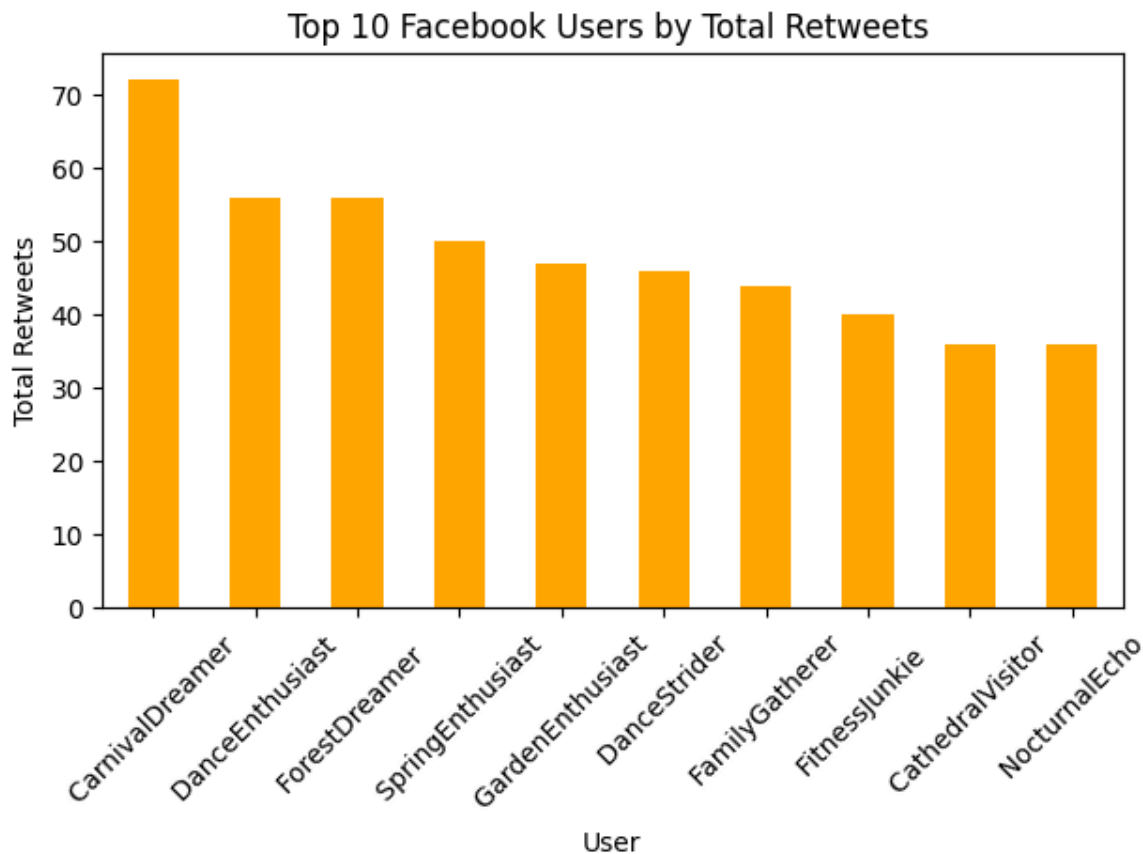


In [204...

```
facebook_data = df[df['Platform'] == 'Facebook']

# Group by User and sum Likes
top_likes_platform_t = facebook_data.groupby('User')['Retweets'].sum().nlargest(10)

# Plotting
top_likes_platform_t.plot(kind='bar', color='orange')
plt.title('Top 10 Facebook Users by Total Retweets ')
plt.xlabel('User')
plt.ylabel('Total Retweets')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [191...

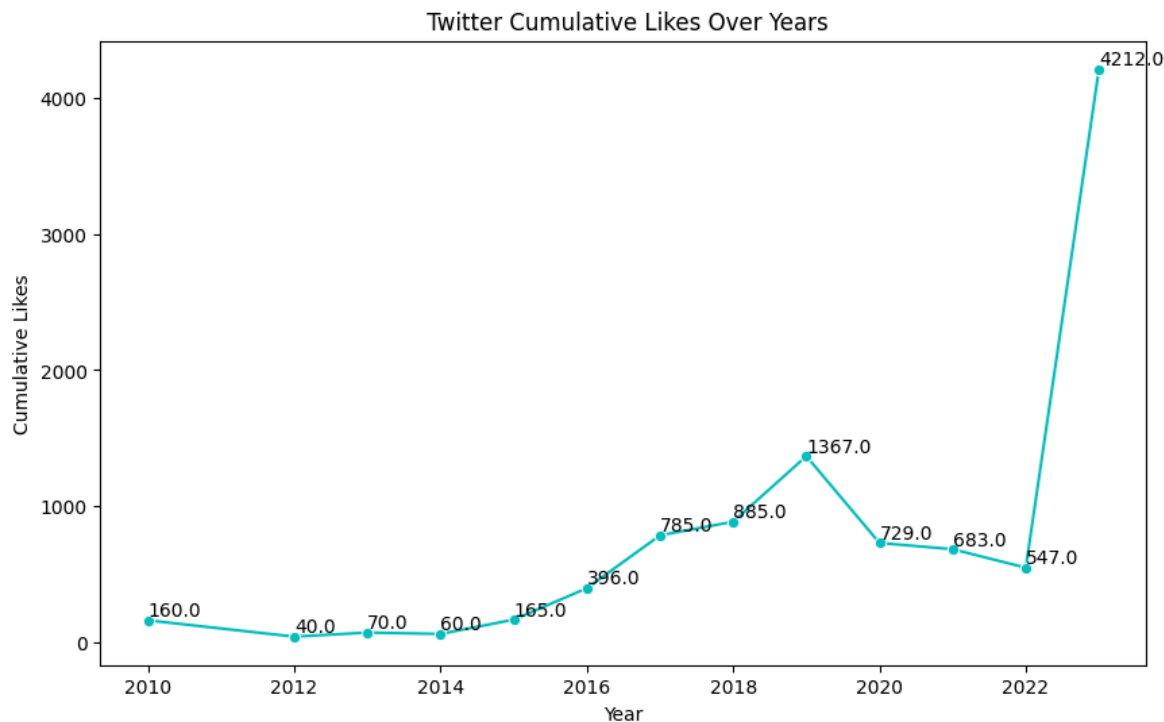
```

twitter=df[df["Platform"] == "Twitter"]
f = twitter.groupby(['Year'])['Likes'].sum().reset_index()
plt.figure(figsize=(10, 6))
sns.lineplot(data=f, x='Year', y='Likes', legend="brief", marker='o', color="c")
for index, value in f.iterrows():
    plt.text(value['Year'], value['Likes'], str(value['Likes']), ha='left', va='bottom')

plt.title(' Twitter Cumulative Likes Over Years')
plt.xlabel('Year')
plt.ylabel('Cumulative Likes')

plt.show()

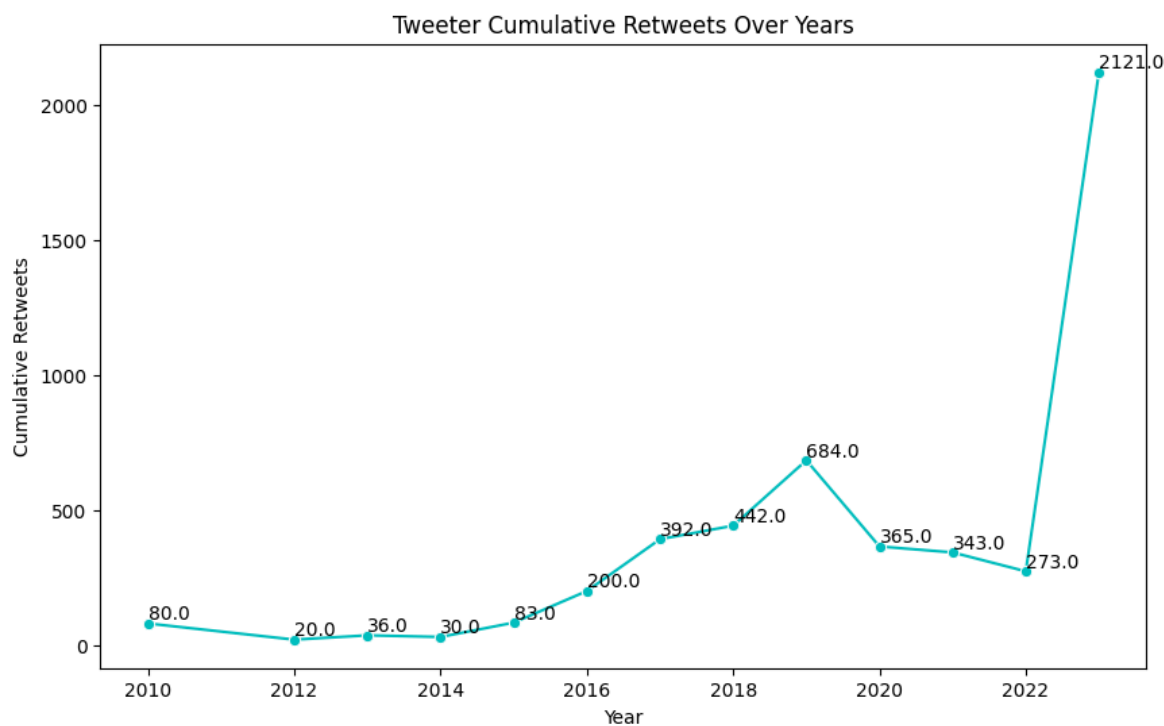
```



In []:

```
In [193... twitter=df[df["Platform"] == "Twitter"]
f = twitter.groupby('Year')['Retweets'].sum().reset_index()
plt.figure(figsize=(10, 6))
sns.lineplot(data=f, x='Year', y='Retweets', marker='o', color="c")
for index, value in f.iterrows():
    plt.text(value['Year'], value['Retweets'], str(value['Retweets']), ha='left')

plt.title('Tweeter Cumulative Retweets Over Years')
plt.xlabel('Year')
plt.ylabel('Cumulative Retweets')
plt.show()
```



In []:

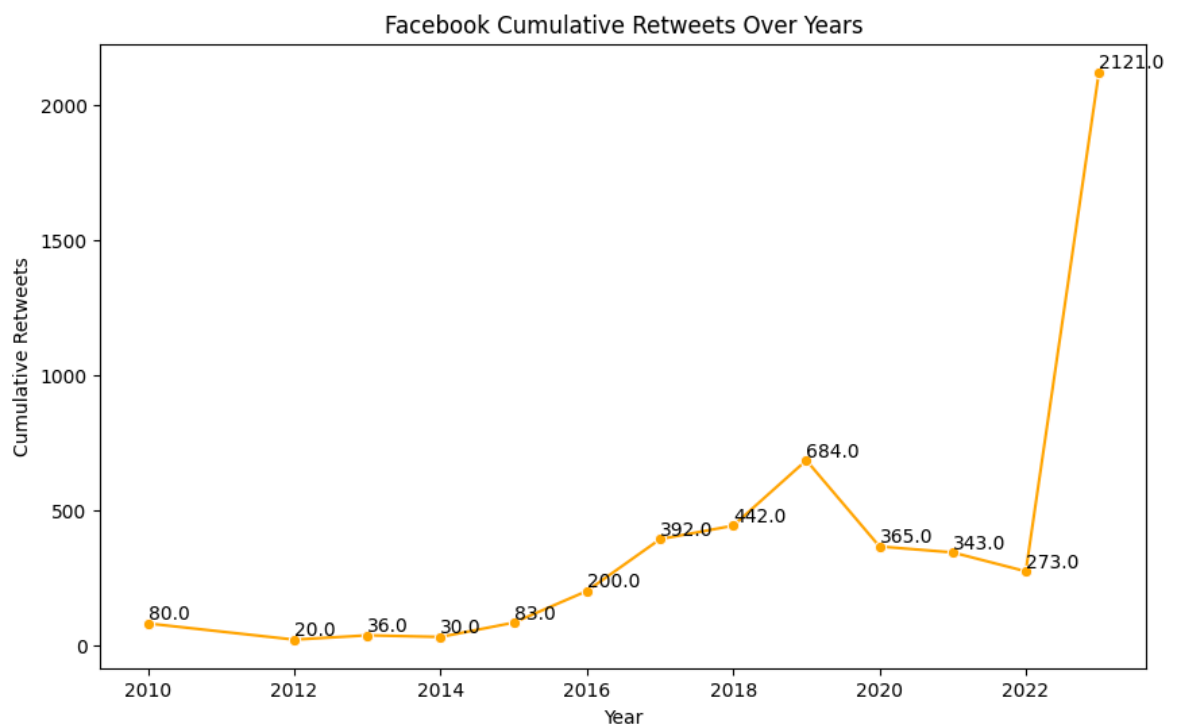
```

In [196... facebook_data = df[df['Platform'] == 'Facebook']
f = Twitter.groupby('Year')['Retweets'].sum().reset_index()
plt.figure(figsize=(10, 6))
sns.lineplot(data=f, x='Year', y='Retweets', marker='o', color="orange")
for index, value in f.iterrows():
    plt.text(value['Year'], value['Retweets'], str(value['Retweets']), ha='left'

plt.title('Facebook Cumulative Retweets Over Years')
plt.xlabel('Year')
plt.ylabel('Cumulative Retweets')

plt.show()

```



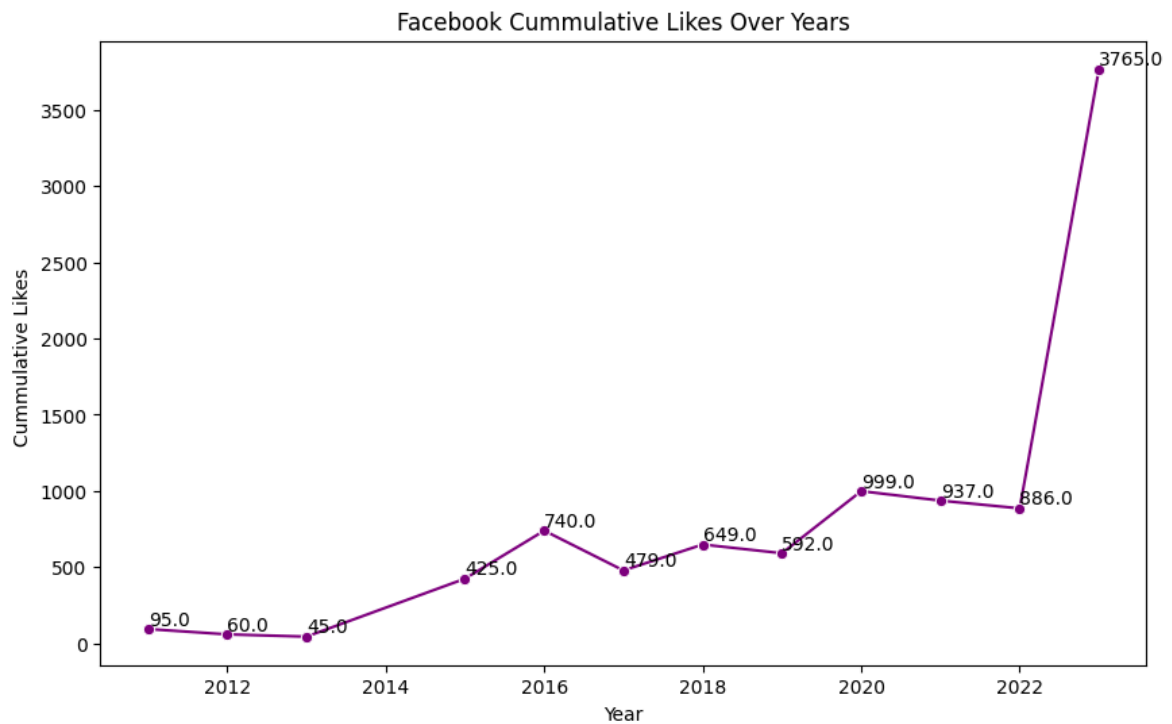
```

In [201... facebook_data = df[df['Platform'] == 'Facebook']
f = Facebook.groupby('Year')['Likes'].sum().reset_index()
plt.figure(figsize=(10, 6))
sns.lineplot(data=f, x='Year', y='Likes', marker='o', color="purple")
for index, value in f.iterrows():
    plt.text(value['Year'], value['Likes'], str(value['Likes']), ha='left', va='

plt.title('Facebook Cumulative Likes Over Years')
plt.xlabel('Year')
plt.ylabel('Cumulative Likes ')

plt.show()

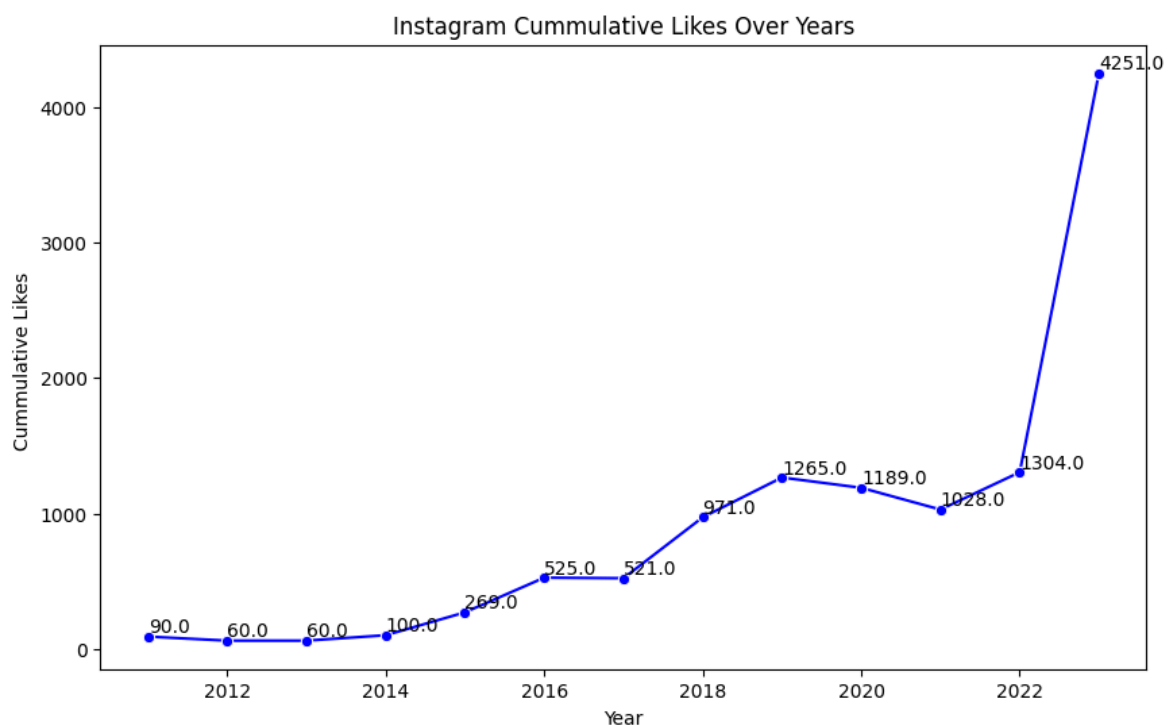
```



```
In [198...
instagram = df[df['Platform'] == 'Instagram']
f = instagram.groupby('Year')['Likes'].sum().reset_index()
plt.figure(figsize=(10, 6))
sns.lineplot(data=f, x='Year', y='Likes', marker='o', color="blue")
for index, value in f.iterrows():
    plt.text(value['Year'], value['Likes'], str(value['Likes']), ha='left', va='

plt.title('Instagram Cummulative Likes Over Years')
plt.xlabel('Year')
plt.ylabel('Cummulative Likes ')

plt.show()
```

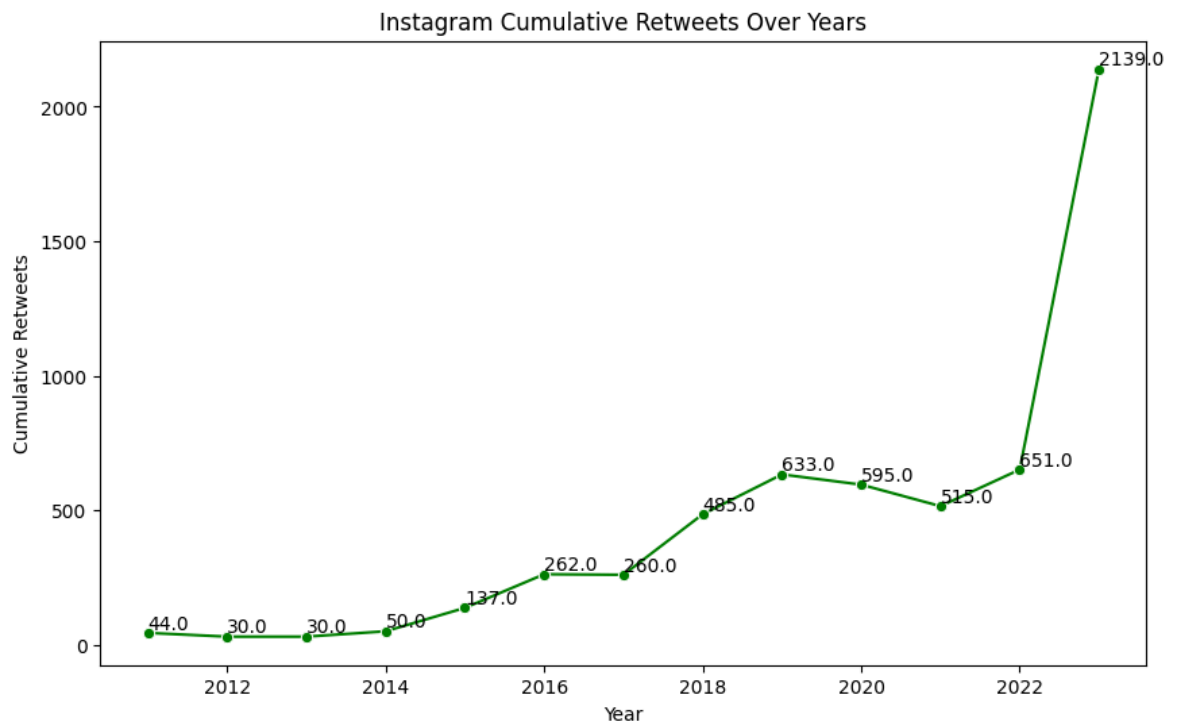


```
In [205...
f = Instagram.groupby('Year')['Retweets'].sum().reset_index()
plt.figure(figsize=(10, 6))
```

```
sns.lineplot(data=f, x='Year', y='Retweets', marker='o', color="green")
for index, value in f.iterrows():
    plt.text(value['Year'], value['Retweets'], str(value['Retweets']), ha='left')

plt.title('Instagram Cumulative Retweets Over Years')
plt.xlabel('Year')
plt.ylabel('Cumulative Retweets')

plt.show()
```



In [184...

```
platforms = ['Twitter', 'Facebook', 'Instagram']
metrics = ['Likes', 'Retweets']

fig, axes = plt.subplots(nrows=3, ncols=2, figsize=(14, 12))
fig.suptitle('Top 10 Users by Likes and Retweets on Each Platform', fontsize=16)

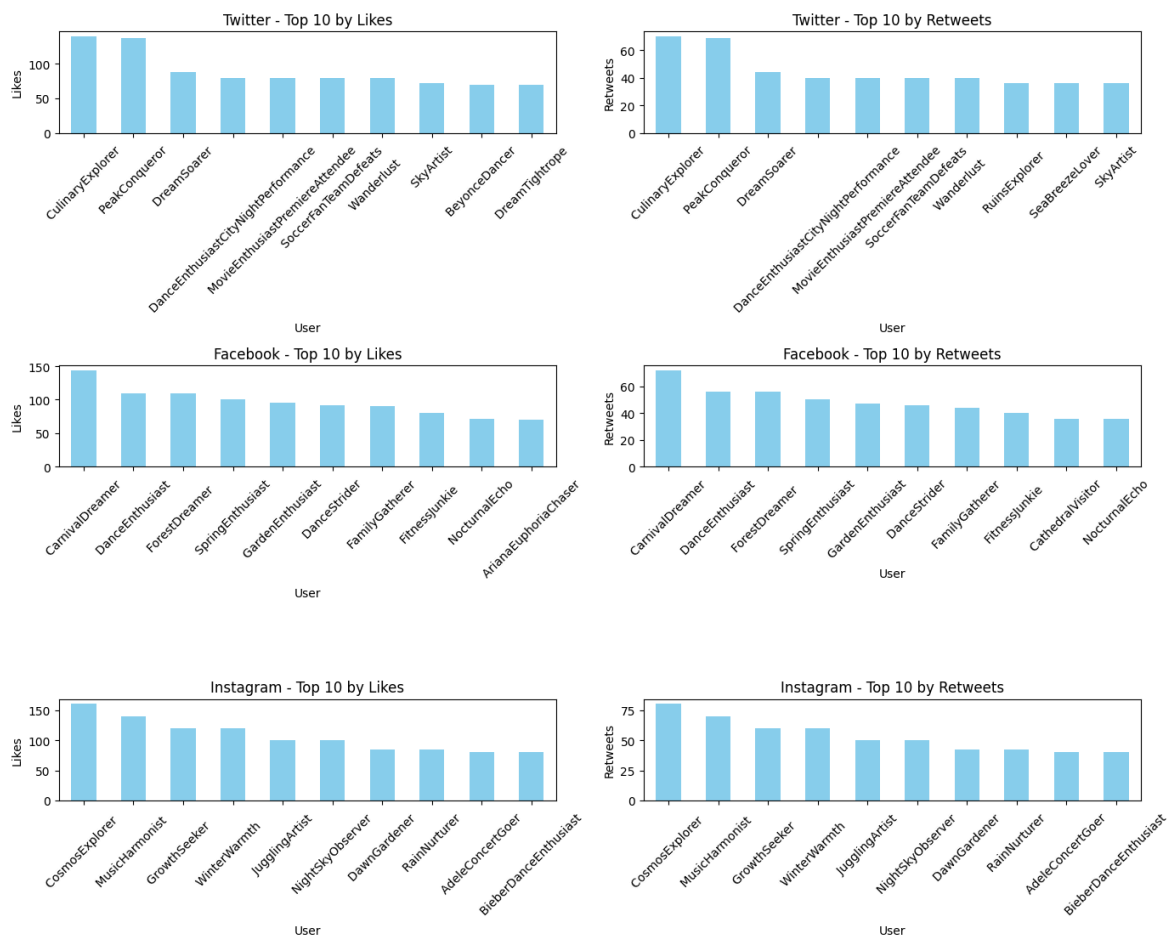
# Loop through platforms and metrics
for i, platform in enumerate(platforms):
    platform_data = df[df['Platform'] == platform]

    for j, metric in enumerate(metrics):
        ax = axes[i, j]
        top_users = platform_data.groupby('User')[metric].sum().nlargest(10)
        top_users.plot(kind='bar', ax=ax, color='skyblue')

        ax.set_title(f'{platform} - Top 10 by {metric}')
        ax.set_xlabel('User')
        ax.set_ylabel(metric)
        ax.tick_params(axis='x', rotation=45)

plt.tight_layout(rect=[0, 0, 1, 0.96]) # Adjust layout to fit the supitle
plt.show()
```

Top 10 Users by Likes and Retweets on Each Platform



In [186...

```
import matplotlib.pyplot as plt

# Platforms to analyze
platforms = ['Twitter', 'Facebook', 'Instagram']
categories = ['Hashtags', 'Sentiment']

# Create subplot grid: 3 rows (platforms) x 2 columns (Hashtags & Sentiments)
fig, axes = plt.subplots(nrows=3, ncols=2, figsize=(14, 12))
fig.suptitle('Top 10 Hashtags and Sentiments by Platform', fontsize=16)

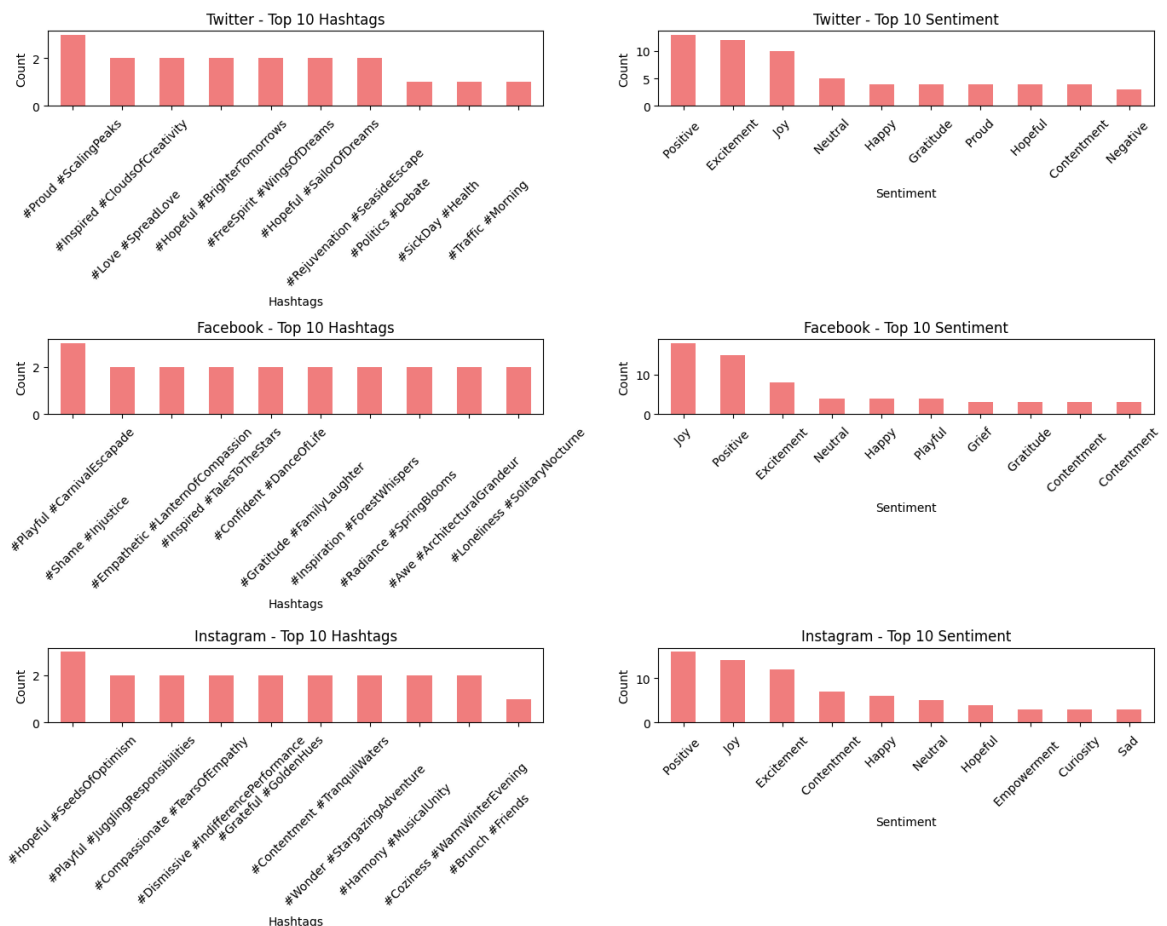
# Loop through each platform and category
for i, platform in enumerate(platforms):
    platform_data = df[df['Platform'] == platform]

    for j, category in enumerate(categories):
        ax = axes[i, j]
        top_values = platform_data[category].value_counts().nlargest(10)
        top_values.plot(kind='bar', ax=ax, color='lightcoral')

        ax.set_title(f'{platform} - Top 10 {category}')
        ax.set_xlabel(category)
        ax.set_ylabel('Count')
        ax.tick_params(axis='x', rotation=45)

plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.show()
```

Top 10 Hashtags and Sentiments by Platform



In [187...

```
import pandas as pd
import matplotlib.pyplot as plt

# Platforms and metrics
platforms = ['Twitter', 'Facebook', 'Instagram']
metrics = ['Likes', 'Retweets']

# Create subplot: 3 rows (platforms) x 2 columns (Likes & Retweets)
fig, axes = plt.subplots(nrows=3, ncols=2, figsize=(14, 12))
fig.suptitle('Cumulative Likes and Retweets Over Years by Platform', fontsize=16)

# Loop through platforms and metrics
for i, platform in enumerate(platforms):
    platform_data = df[df['Platform'] == platform]

    for j, metric in enumerate(metrics):
        ax = axes[i, j]

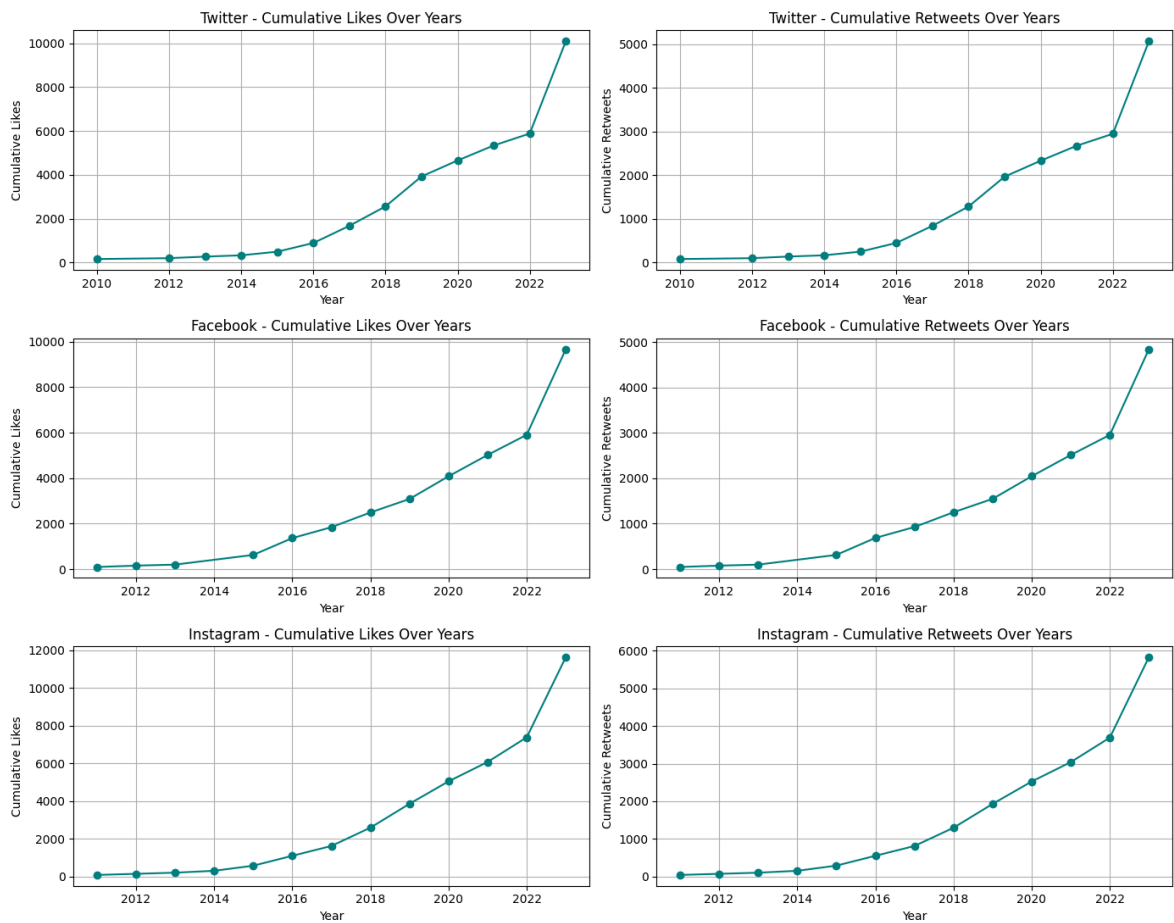
        # Group by year and calculate cumulative sum
        yearly_data = (
            platform_data.groupby('Year')[metric]
            .sum()
            .cumsum() # Cumulative sum over years
        )

        yearly_data.plot(kind='line', marker='o', ax=ax, color='teal')
        ax.set_title(f'{platform} - Cumulative {metric} Over Years')
        ax.set_xlabel('Year')
        ax.set_ylabel(f'Cumulative {metric}')
```

```
ax.grid(True)

plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.show()
```

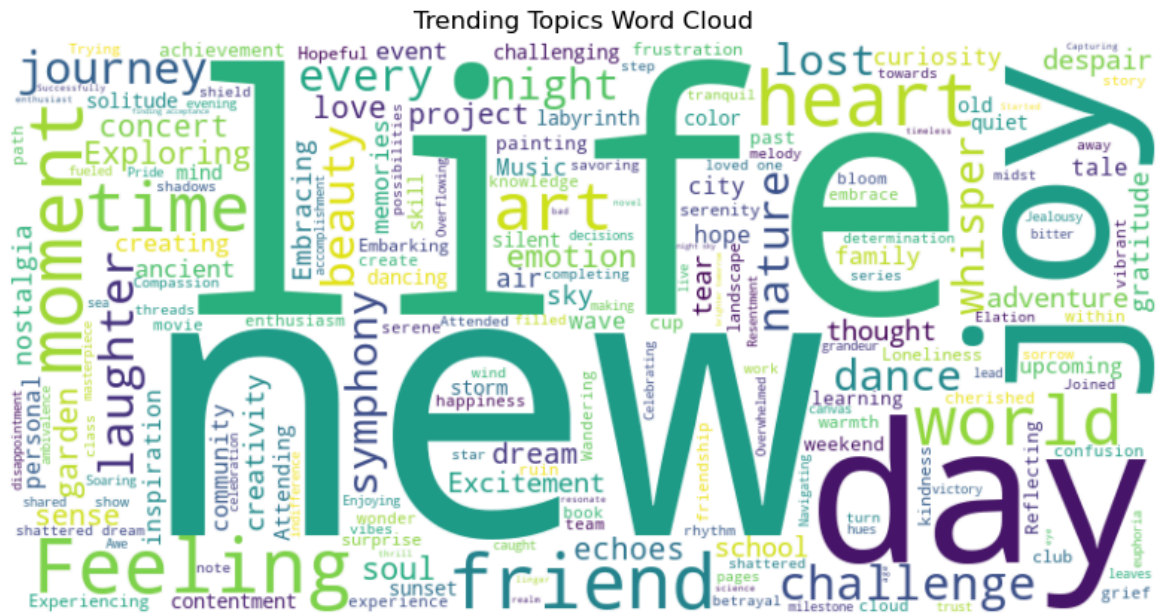
Cumulative Likes and Retweets Over Years by Platform



In [170...

```
# Word Cloud for Trending Topics
all_words = ' '.join(df['Text'])
wordcloud = WordCloud(width=800, height=400, background_color='white').generate(

plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis('off')
plt.title("Trending Topics Word Cloud")
plt.show()
```



Thank You !

For Visiting